A PROJECT REPORT

ON

“Virtual Platform Creation and Security Implementation Report”

Name :- Nitish Kumar
ID :- 445-11999
Roll: B-43
Batch :- BCA PPU Batch-2 3rd year

Submitted to: Respected Murli Sir

1. Introduction

Application Packaging is an important process used in modern IT environments to deploy and manage software across multiple computers efficiently. In large organizations, installing applications manually on each computer is time-consuming and error-prone. Application packaging solves this problem by creating standardized installer packages that can be deployed automatically.

In this project, a virtual machine environment is created to simulate a real enterprise setup. The system is prepared, applications are installed, monitored, packaged, tested, and deployed using packaging tools.
The goal of this project is to understand the complete lifecycle of application packaging including environment preparation, packaging, testing, and deployment.

2. Objectives of the Project

The main objectives of this project are:

- To understand virtualization and virtual machine setup
- To prepare a clean operating system environment
- To install and configure required software
- To capture application installation for packaging
- To create MSI based application packages
- To test the application package for proper installation
- To understand enterprise software deployment process

3. Tools and Technologies Used

The following tools and technologies were used in the project:
VMware Workstation / VirtualBox – For creating virtual machines

- Windows 10 / Windows 11 – Operating system environment
- MSI Packaging Tools – For creating application installer packages
- Registry Editor – For security and configuration changes
- Winget Package Manager – For updating installed applications

4. System Requirements

Hardware Requirements:-

Processor: Intel i3 or higher
RAM: Minimum 8GB
Storage: Minimum 100GB free disk space
Software Requirements:-

VMware Workstation / VirtualBox
Windows 10 or Windows 11 ISO
Application Packaging Tool
Internet Connection

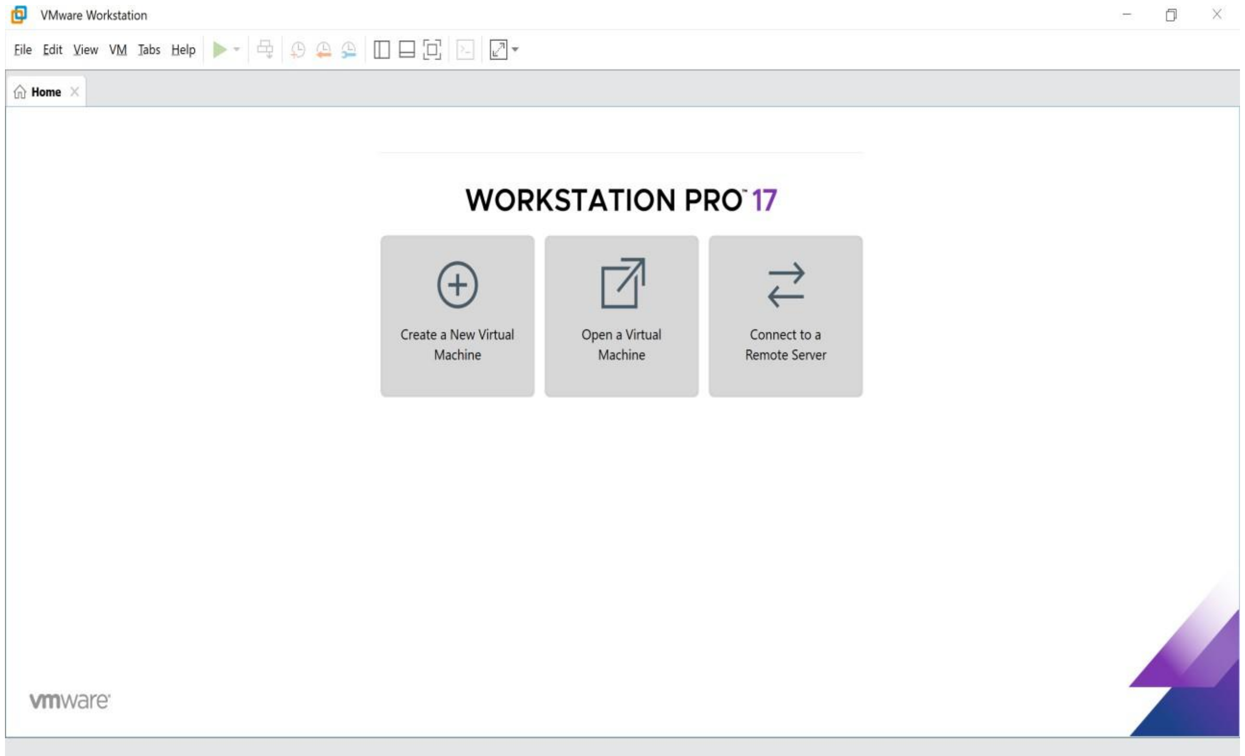
Unit -1 VM Installation & Configuration

1. Choose a Virtualisation Platform

Virtualization platform allows us to run multiple operating systems on a single physical computer. In this project VMware Workstation Player was used to create and manage the virtual machine environment.

Steps:-

- Download VMware Workstation Player from official website.
- Install the software on the host computer.
- Launch VMware Workstation Player.



2. Create a New Virtual Machine

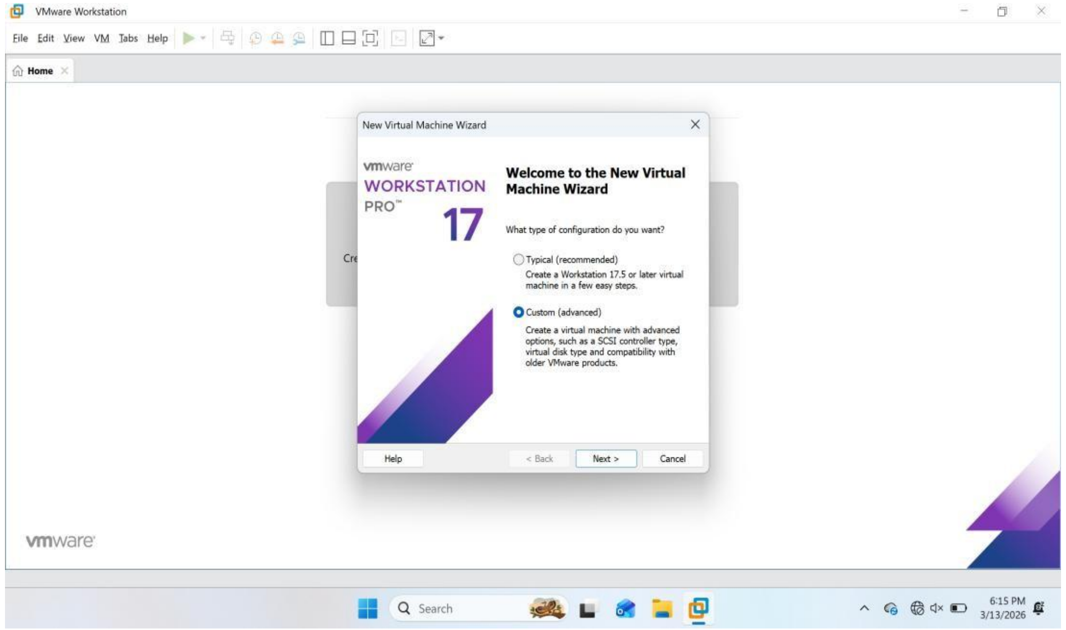
A virtual machine was created to install the Windows operating system.

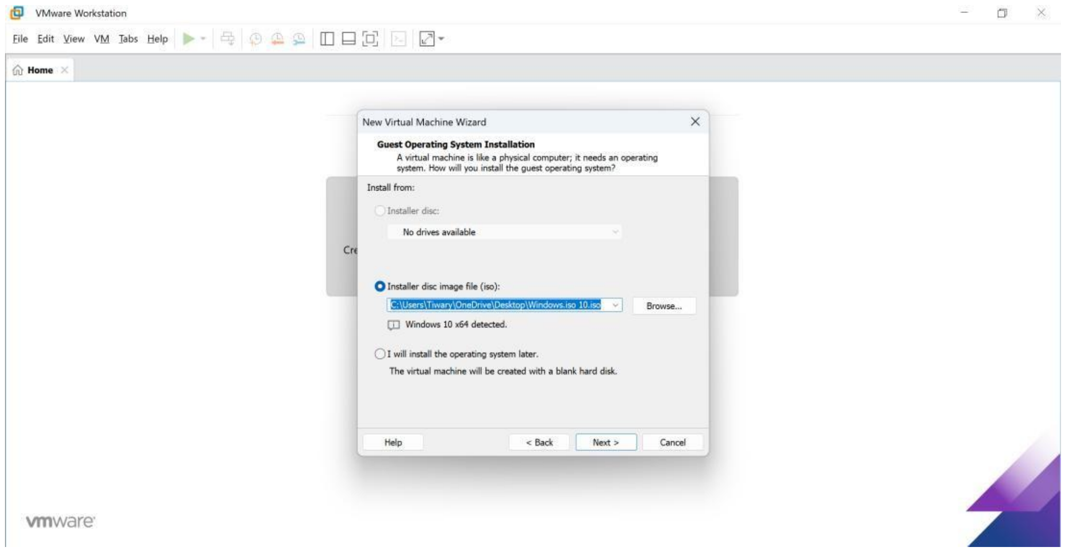
Steps:

- Open VMware Workstation Player.
- Click Create a New Virtual Machine.
- Select Installer Disc Image File (ISO).
- Choose Windows ISO file.
- Enter virtual machine name and location.
- Allocate CPU, RAM and disk space.

Configuration used in this project:

RAM: 4 GB
CPU: 2 Cores
Disk Size: 50 GB



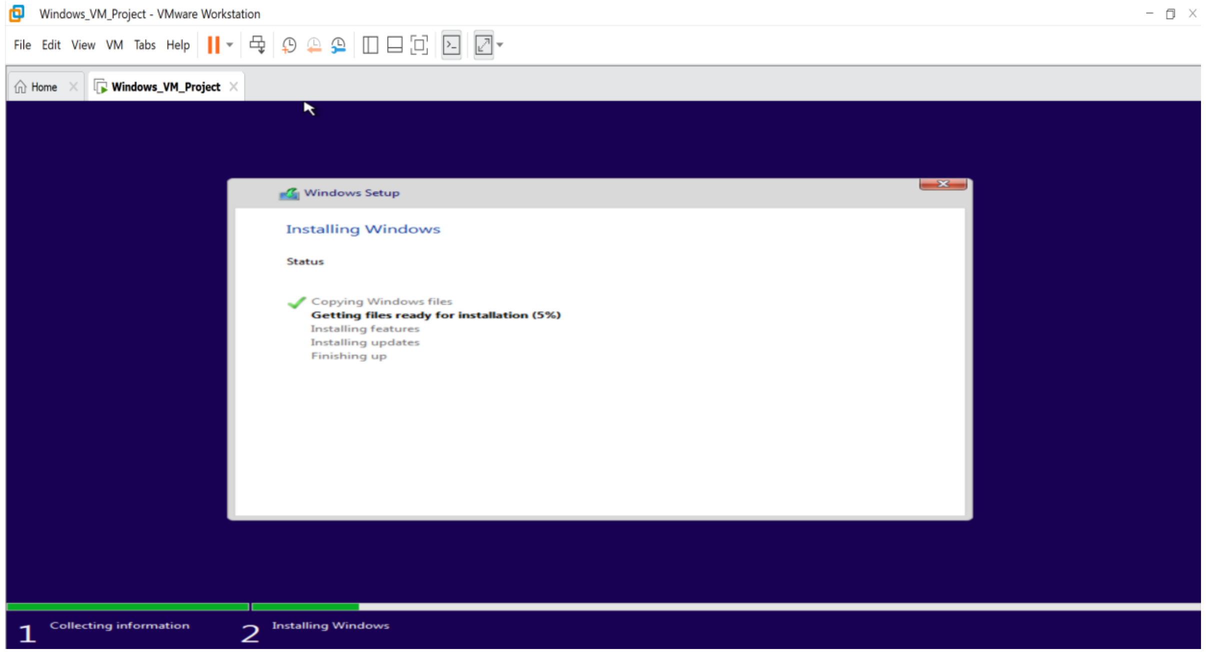


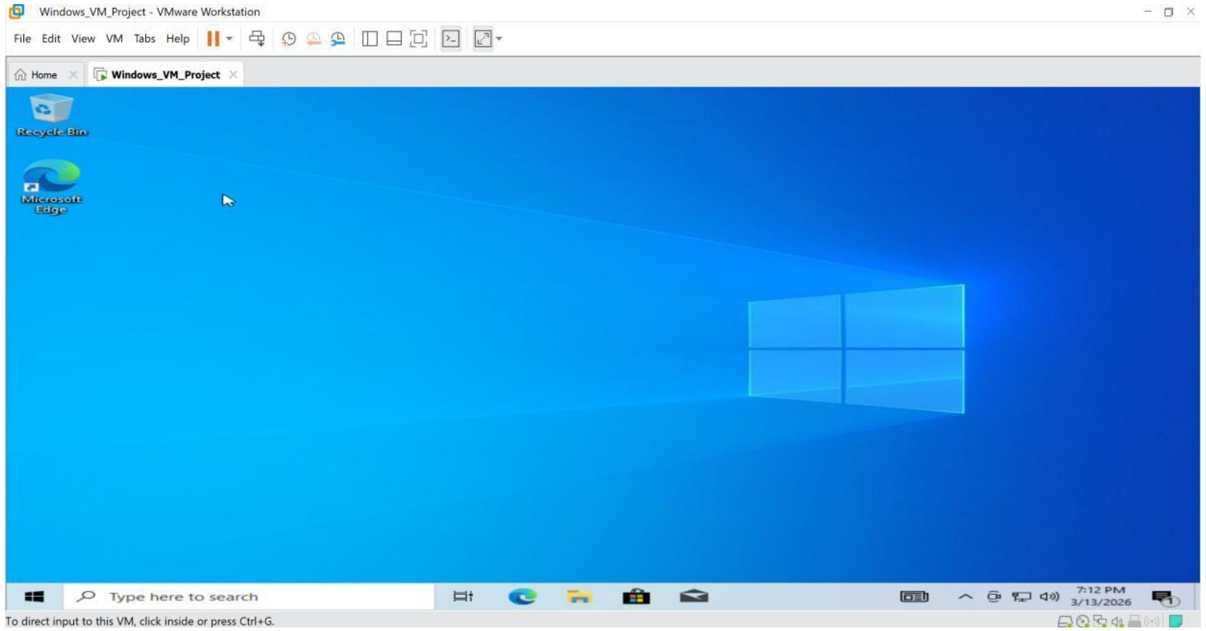
3. Install Windows 10

The Windows operating system was installed inside the virtual machine using an ISO file.

Steps:-

- Start the virtual machine.
- Windows setup screen appears.
- Select language and keyboard layout.
- Click Install Now.
- Accept the license agreement.
- Choose Custom Installation.
- Select disk and install Windows.





4. Apply Updates

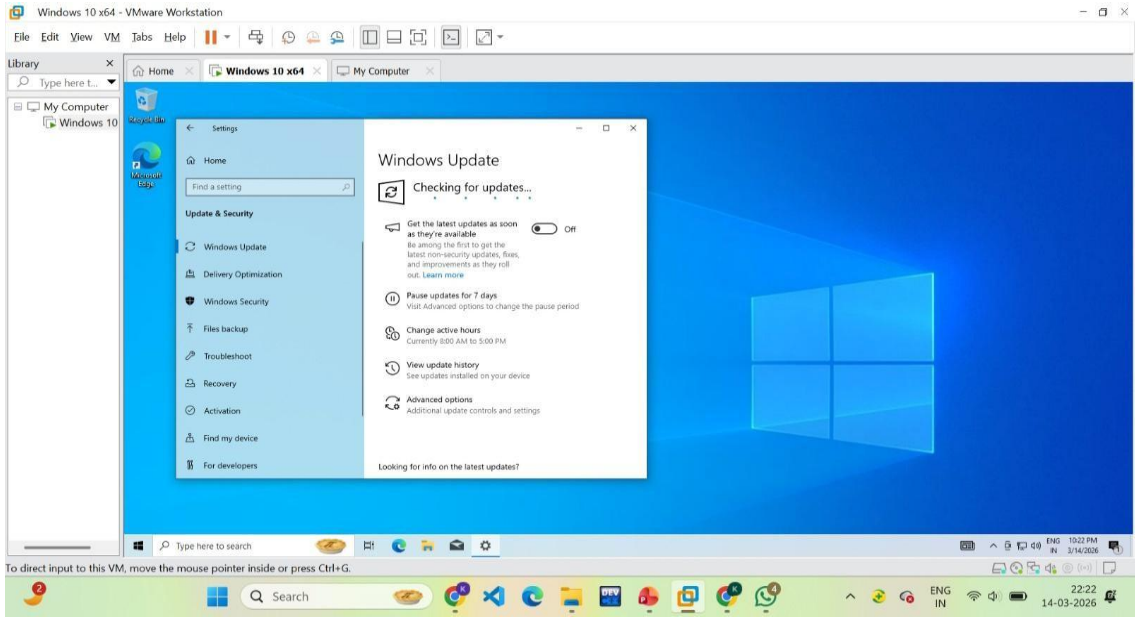
Windows updates were installed to ensure the operating system is up to date and secure.

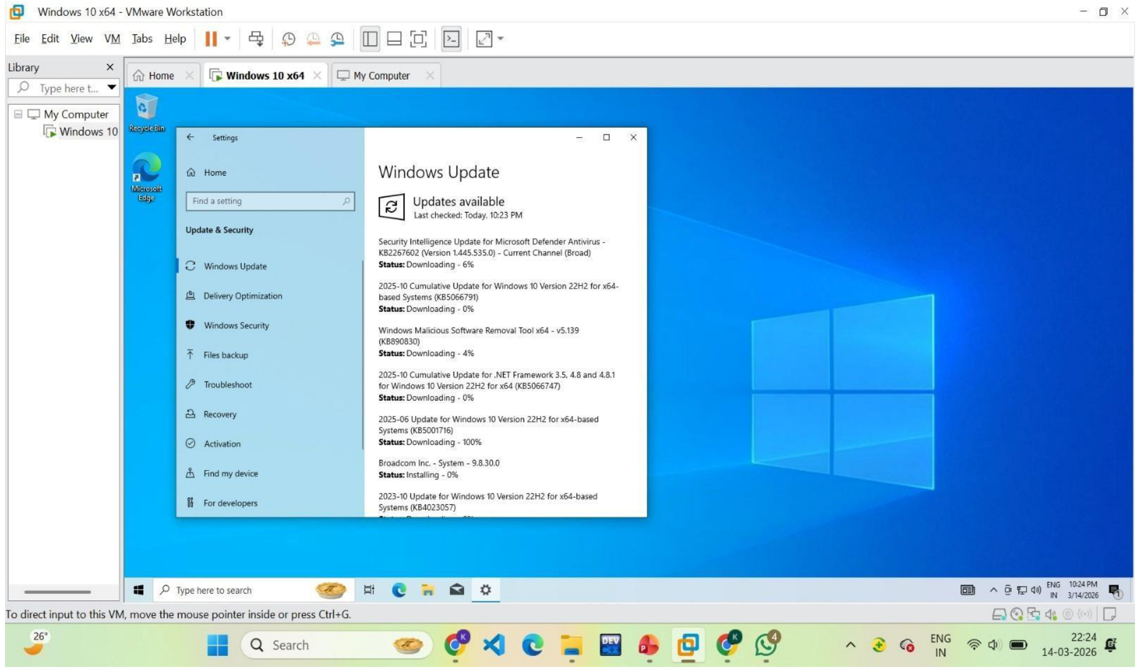
Steps:-

- Open Settings.
- Navigate to Windows Update.
- Click Check for Updates.
- Install all available updates.
- Applications were also updated using Winget.

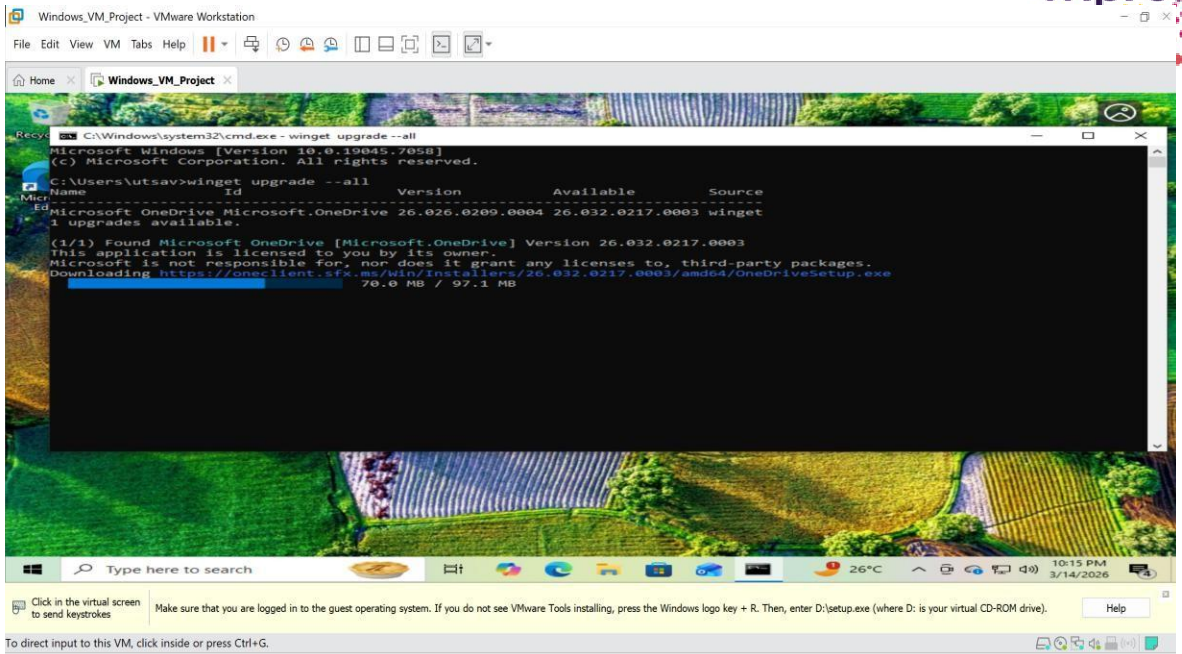
Command used:

winget upgrade --all





5. **Updating Applications using Winget**

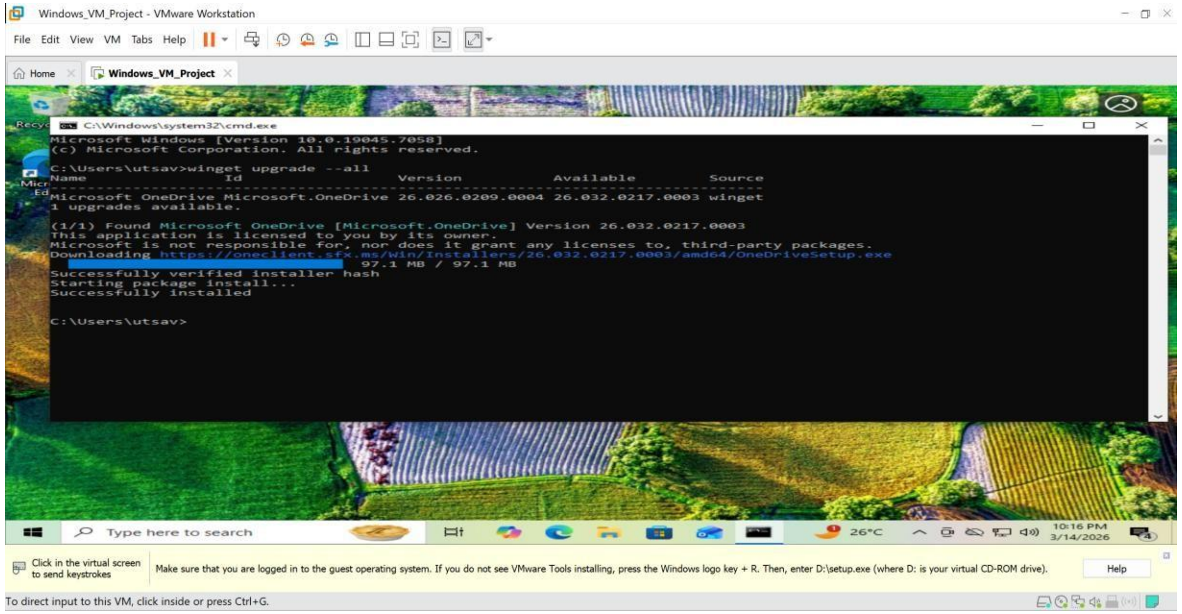


Winget is the Windows Package Manager used to update applications from the command line.

Steps:-

- Open Command Prompt as Administrator.
- Run the following command:

  winget upgrade --all
- The system checks for available updates and installs them automatically.

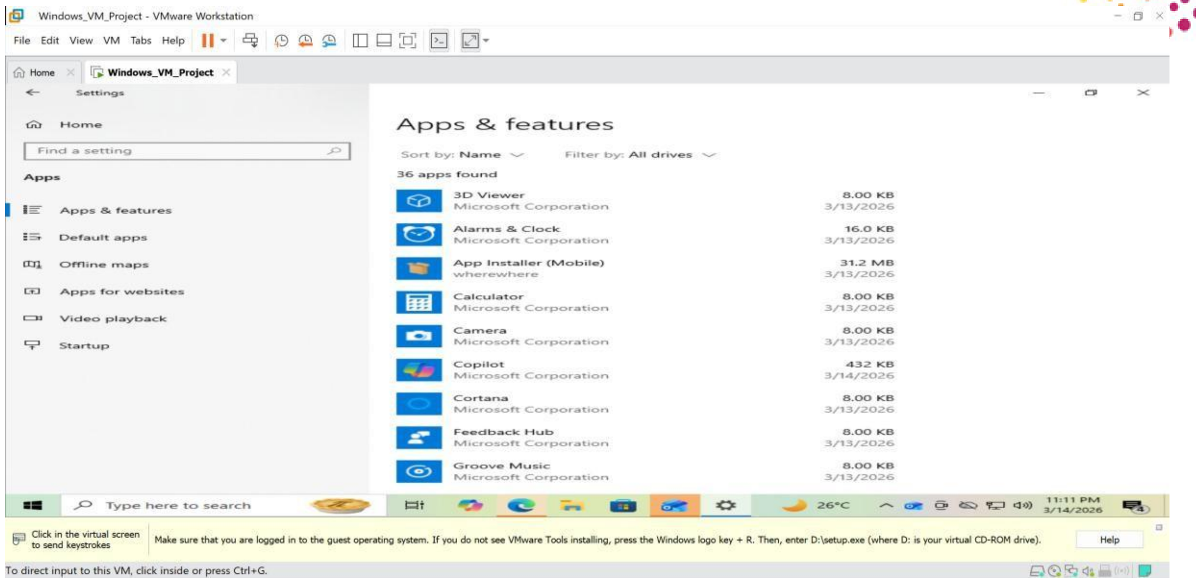


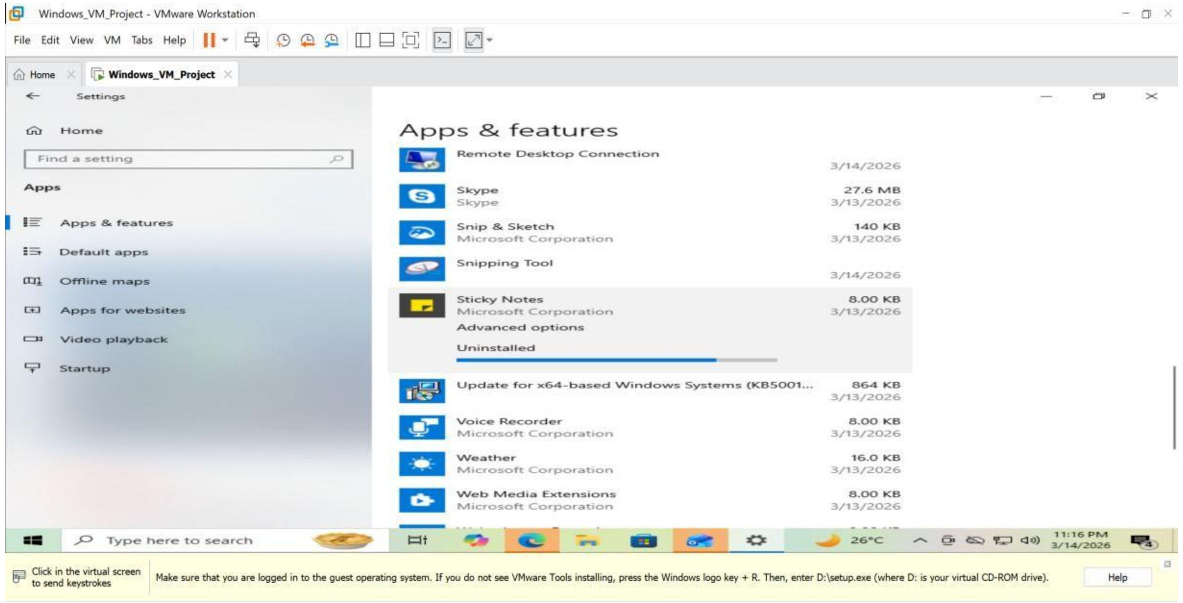
6. Standardizing the Environment

Unnecessary applications were removed to maintain a clean environment for testing and packaging.

Steps:-

- Open Settings.
- Navigate to Apps → Installed Apps.
- Identify unnecessary applications.
- Uninstall unwanted software



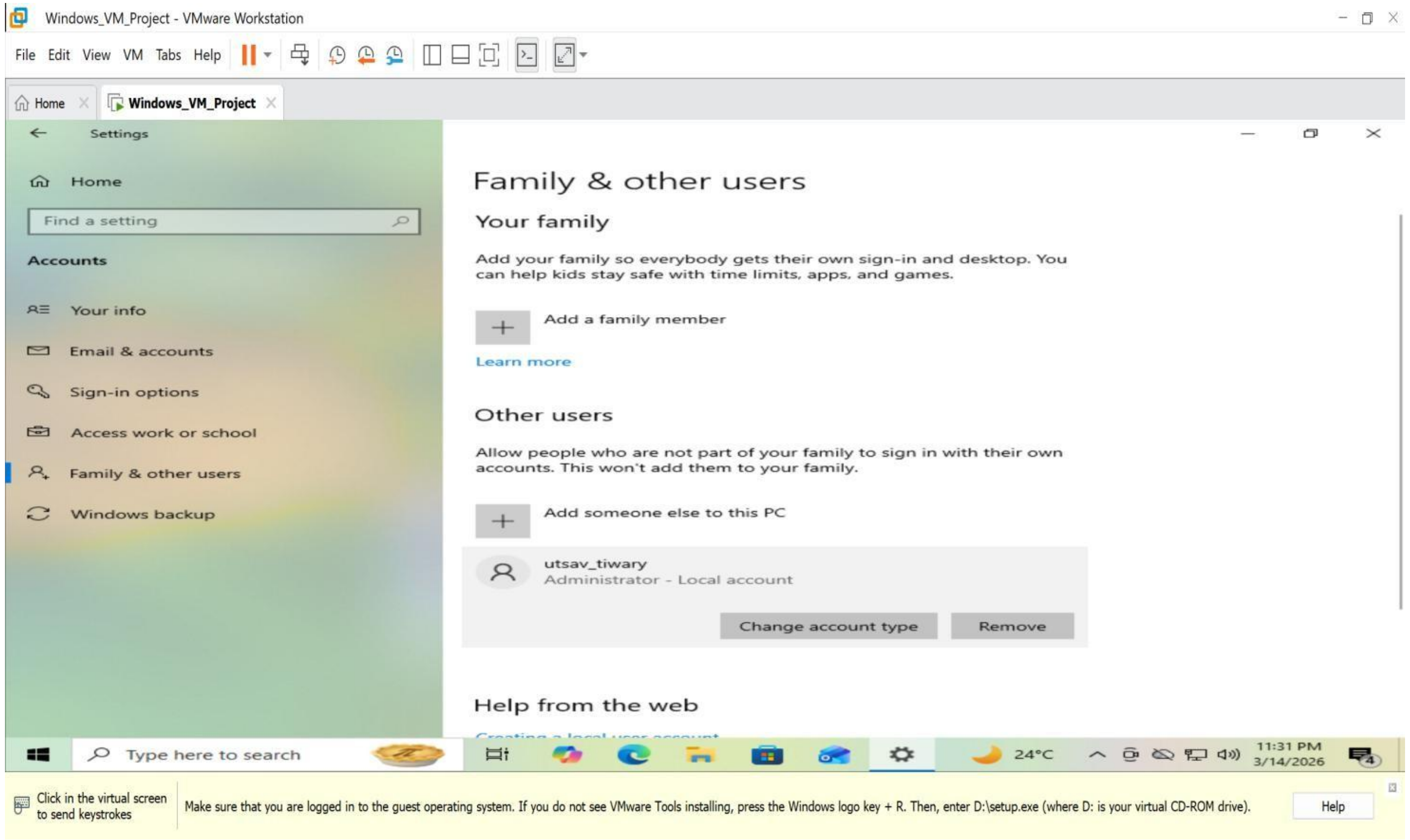


7. User Management

A new user account was created and administrative privileges were assigned.

Steps:-

- Open Settings.
- Go to Accounts → Other Users.
- Click Add Account.
- Create a new user account.
- Assign administrator privileges.



8. Security with Registry Editor

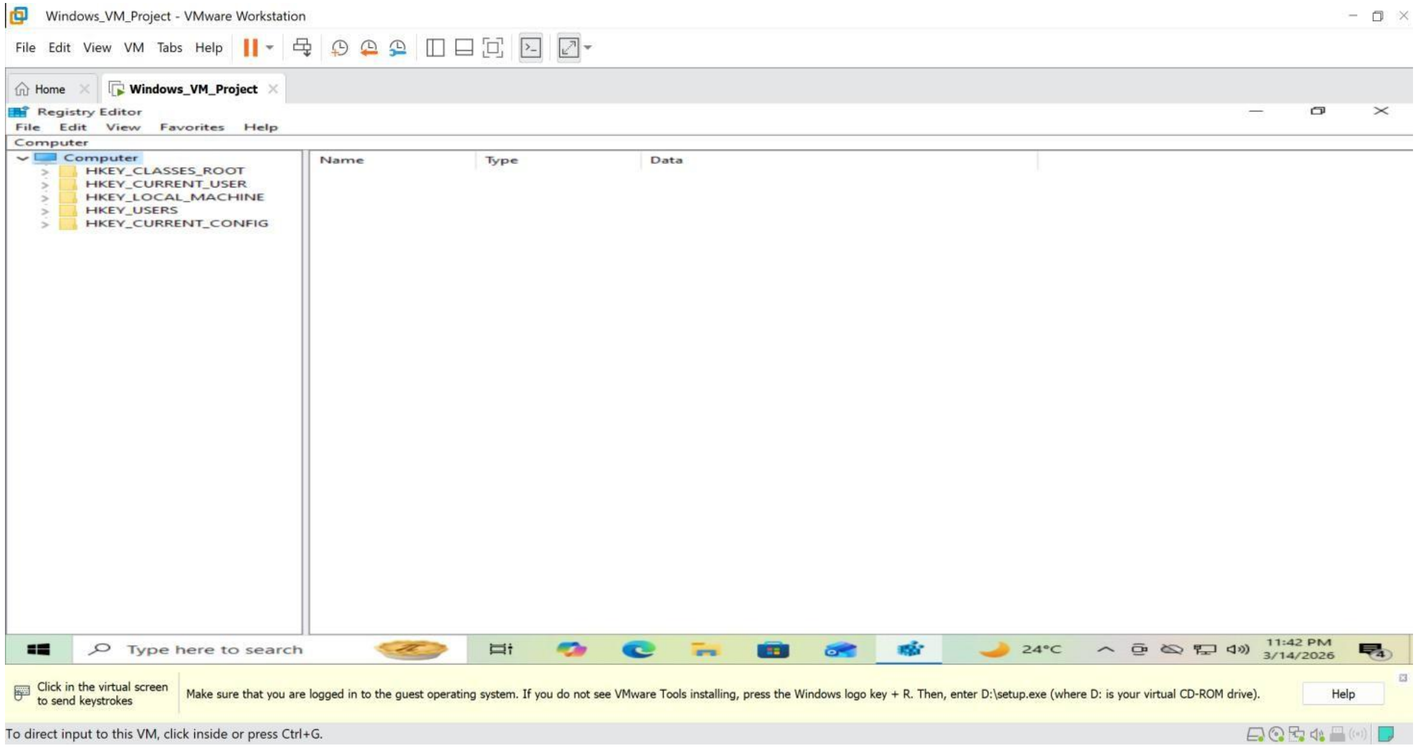
Registry Editor allows administrators to manage system configuration settings.

Steps

- Press Windows + R.
- Type the following command:

  regedit

- Press Enter to open Registry Editor.



Unit 2 :- Snapshot Creation For Rollback

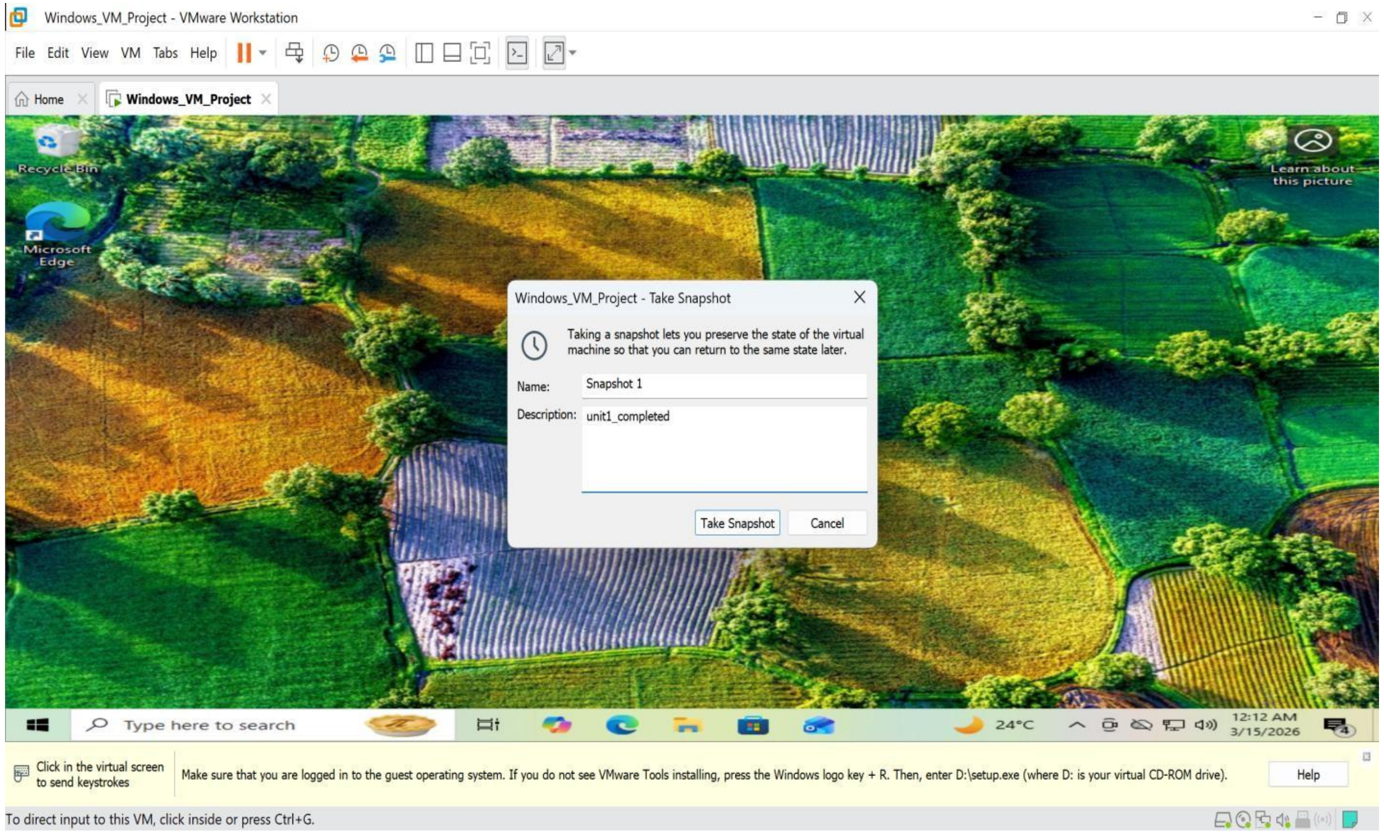
Snapshots capture the current state of the virtual machine so it can be restored later.

Steps to Create Snapshot

- Open VMware Workstation Player.
- Select the virtual machine.
- Click:

VM → Snapshot → Take Snapshot

Provide a name for the snapshot

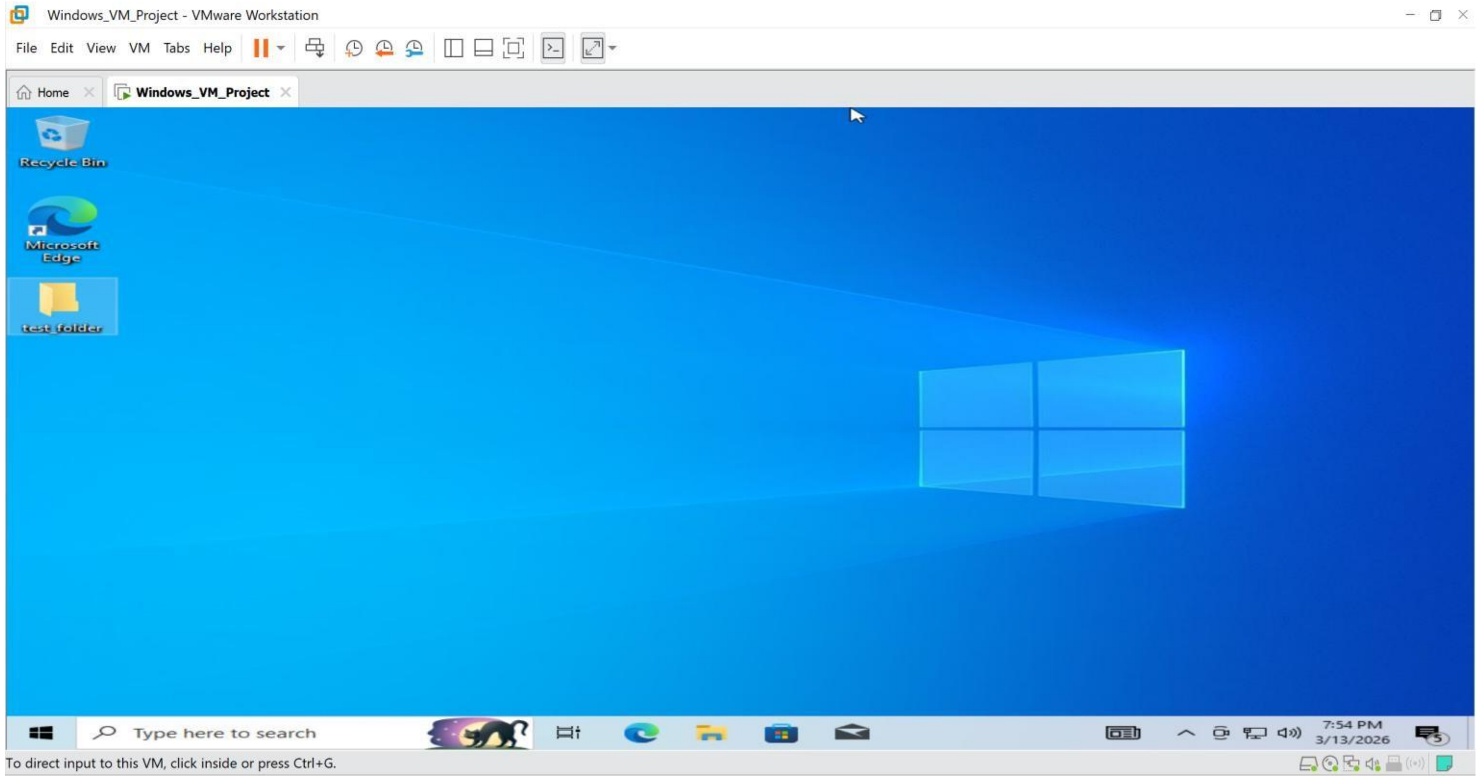


Making Changes in Virtual Machine

After creating the snapshot, some changes were made to the system.

Example changes:

- Creating a folder on the desktop
- Installing a new application
- Changing system settings



New folder created on desktop

Snapshot Rollback

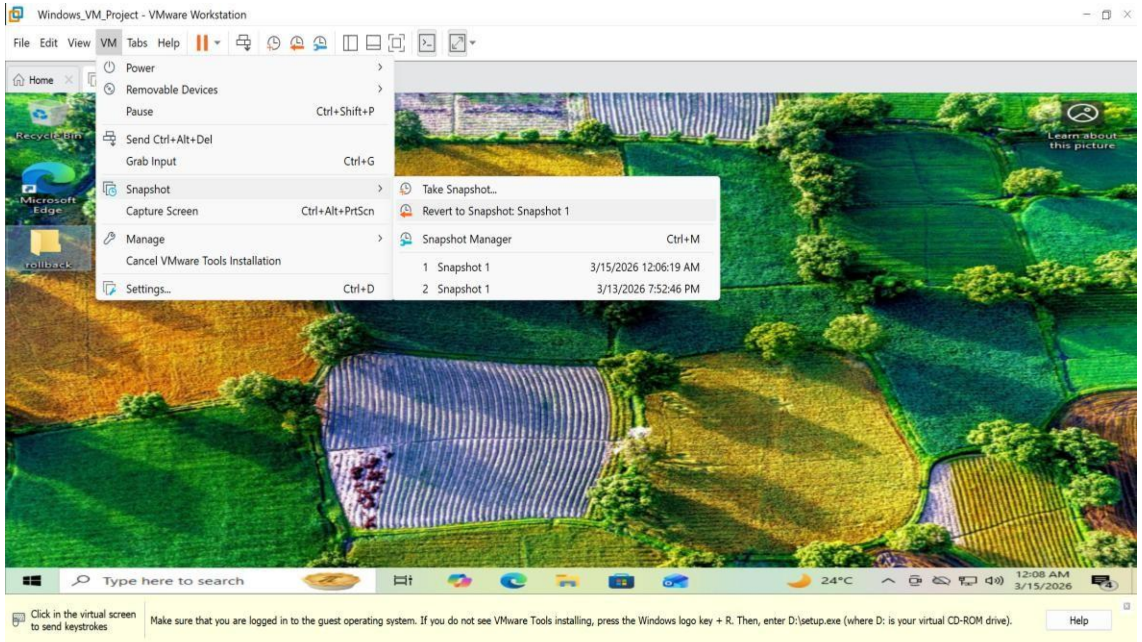
The system was reverted to the previous state using snapshot functionality.

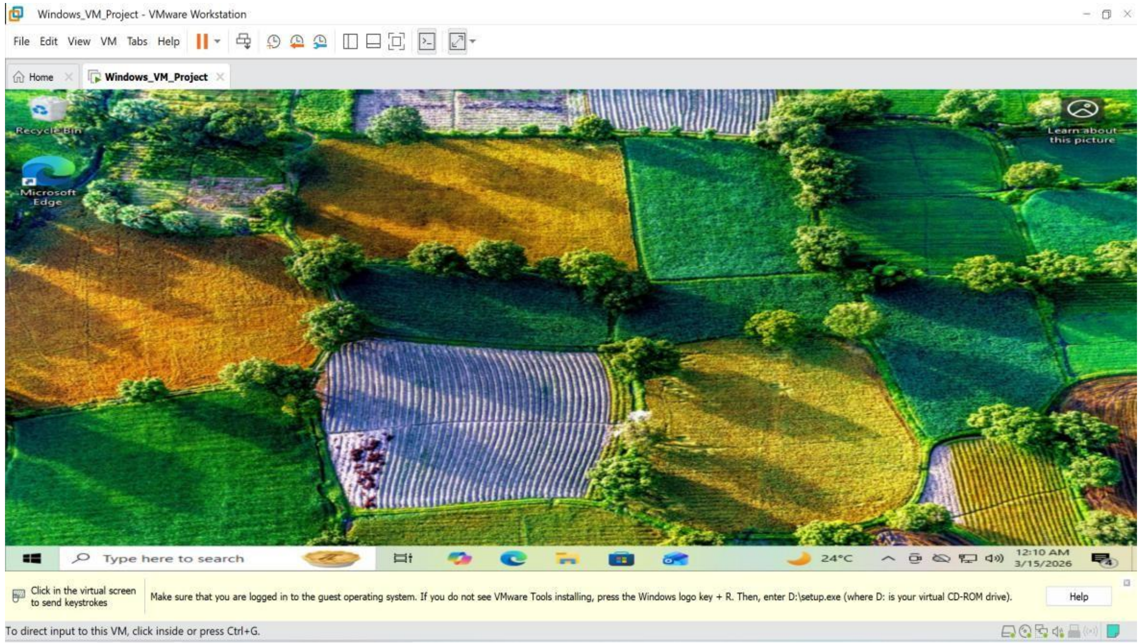
Steps

- Open VMware menu.
- Click:

  VM → Snapshot → Revert to Snapshot

- The system returned to the previous saved state.



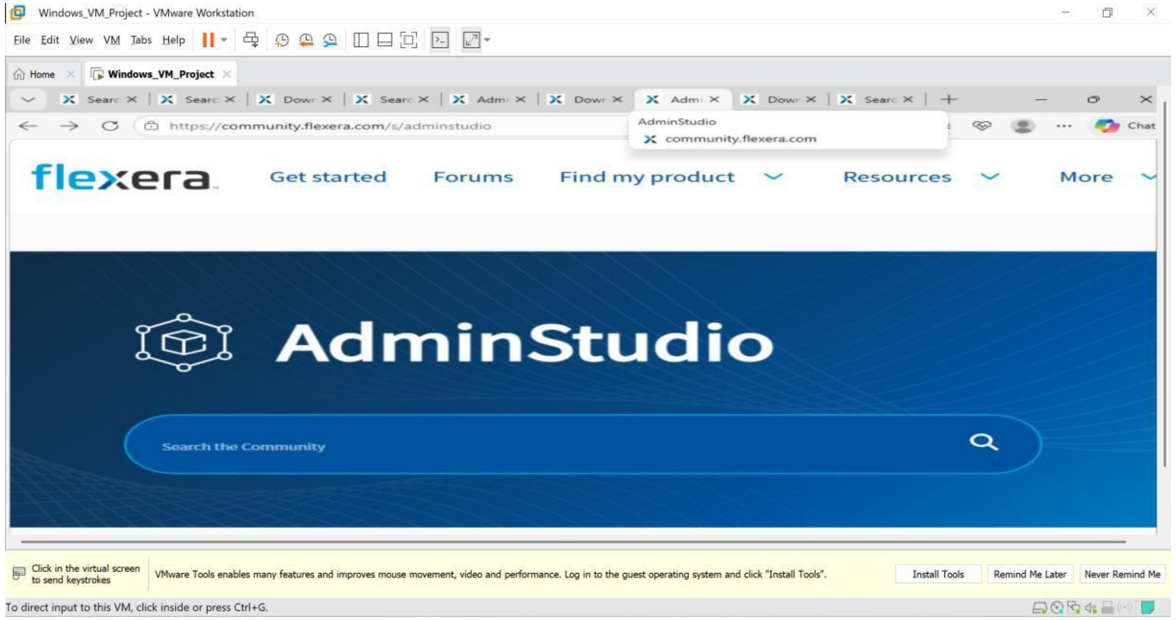


Unit 3 – Install Packaging Tools

Step 1: Download AdminStudio

AdminStudio trial version was downloaded from the official Flexera community website. The download link provided in the project document was used to obtain the installation setup of AdminStudio.

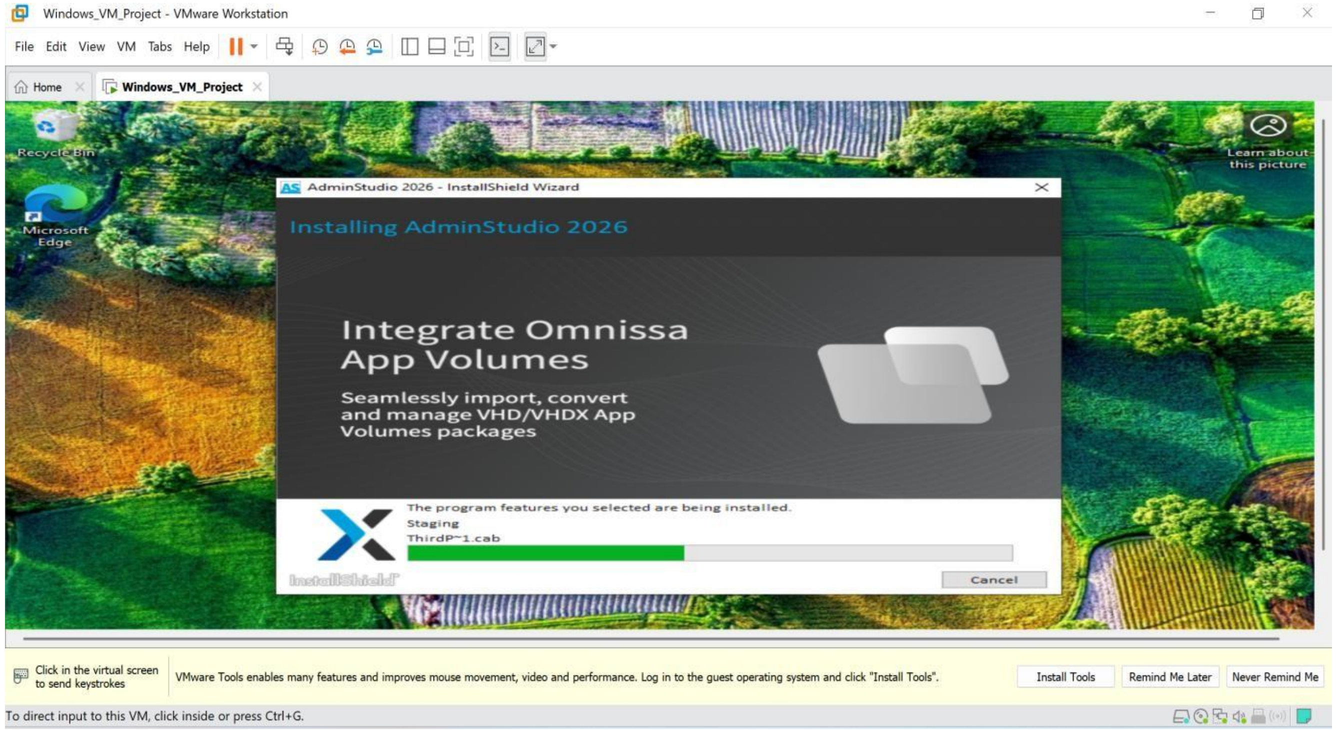
Link used for download:
https://community.flexera.com/s/article/download-adminstudio



Step 2: Install AdminStudio on Virtual Machine

The downloaded AdminStudio setup file was opened inside the Windows virtual machine. The installation wizard was followed to complete the installation process.
Installation steps performed:

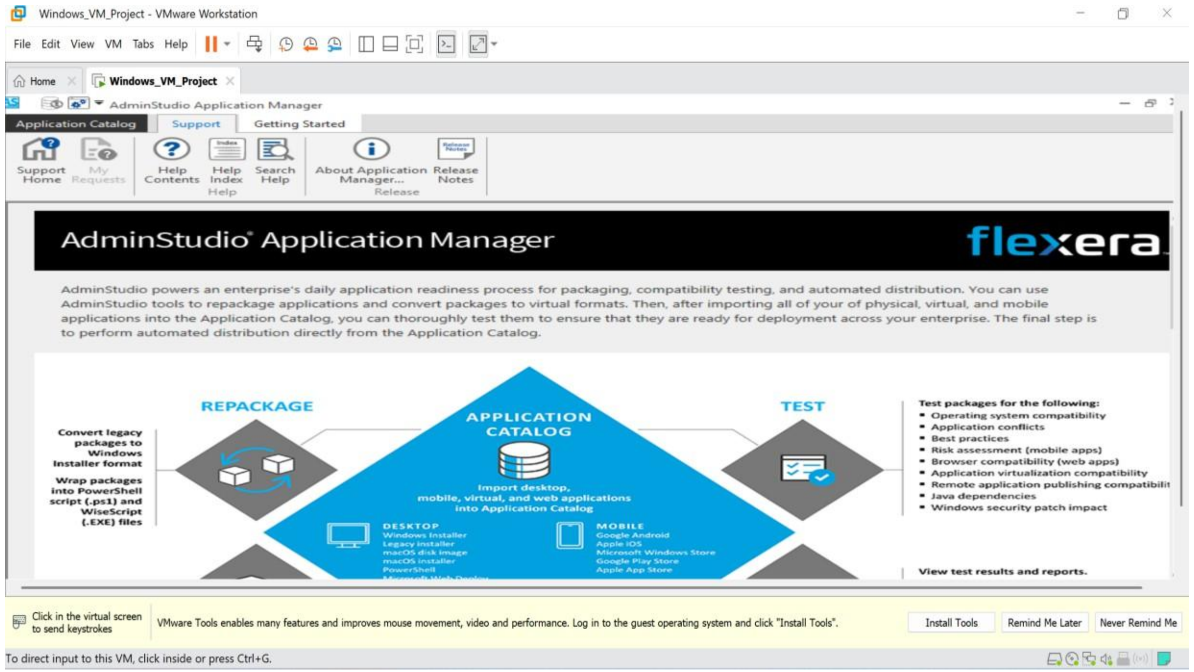
- Opened the AdminStudio setup file.
- Clicked on Next to start the installation wizard.
- Accepted the License Agreement.
- Selected the Typical Installation option.
- Clicked Install to start the installation process.
- Waited for the installation to complete.
- Clicked Finish to complete the setup.

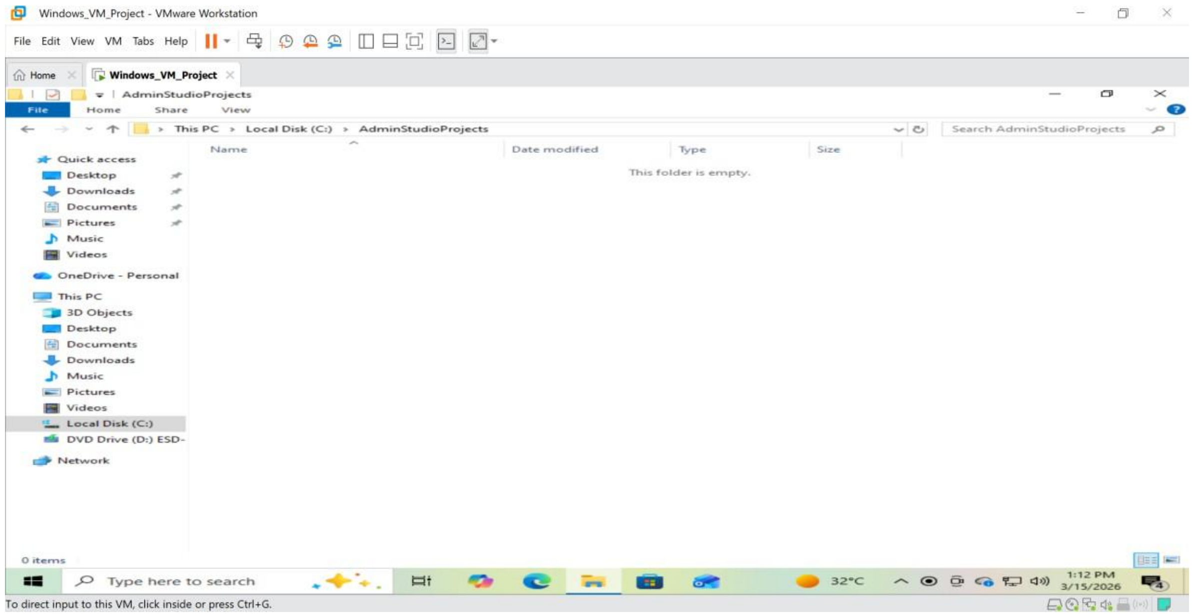


Step 3: Configure Basic Settings

After installation, AdminStudio was launched from the Start menu. The trial license option “Continue to Evaluate AdminStudio” was selected to activate the evaluation version. A default project directory was created to store packaging projects.
Project directory created:

C:\AdminStudioProjects
This directory will be used to store and manage application packaging projects during the project work.

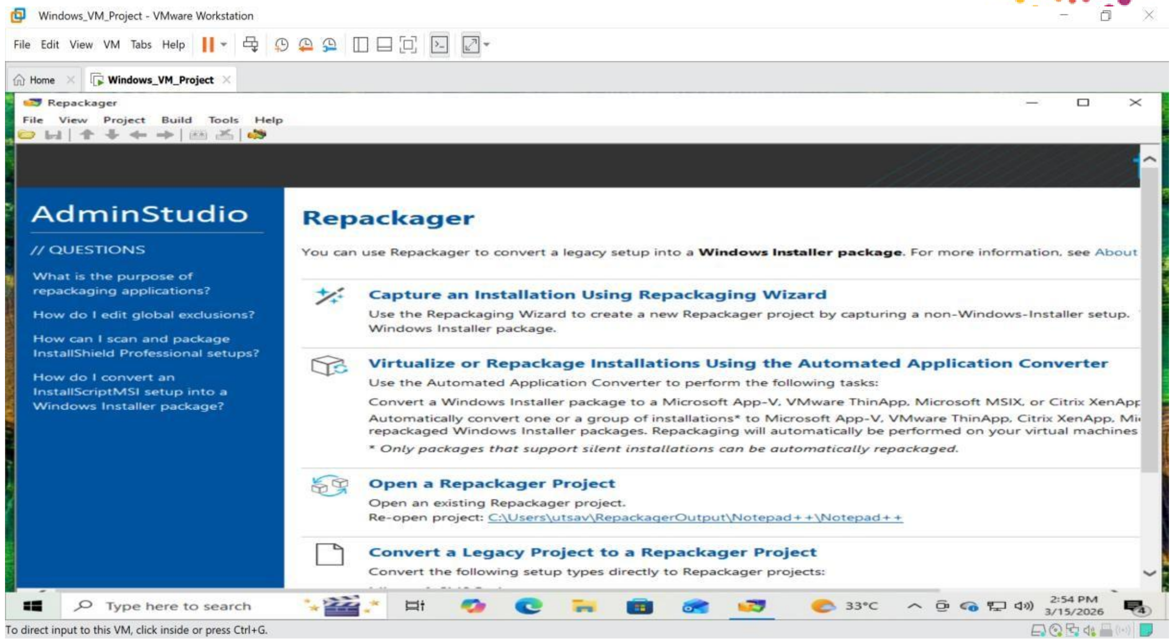




Unit 4 – Application Capture and Msi Package Creation

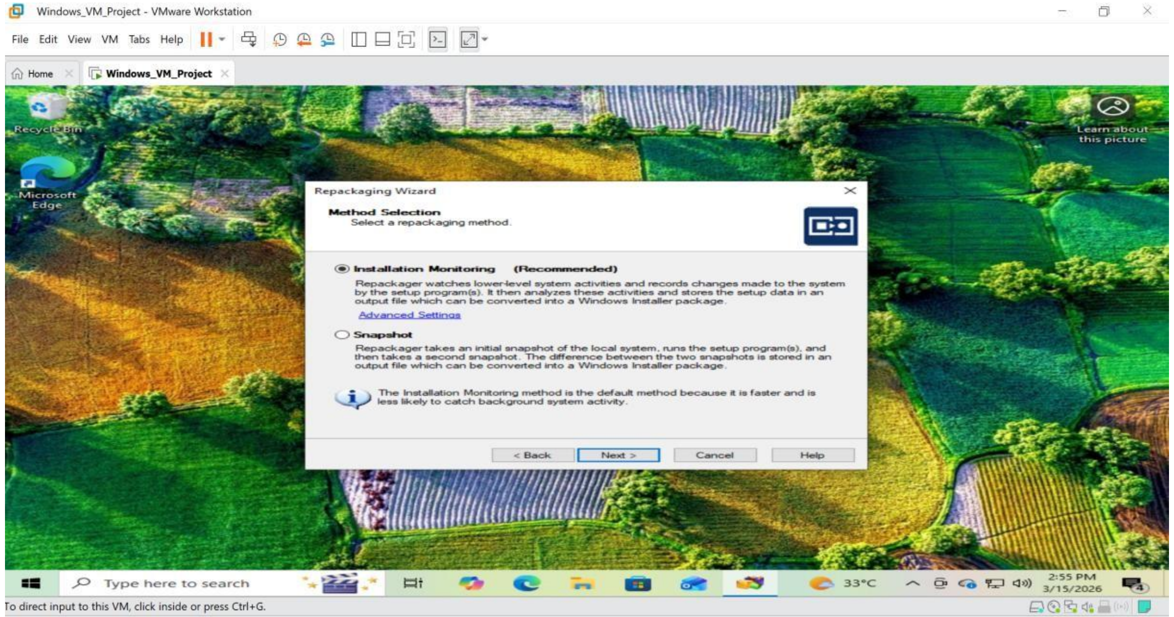
Step 1 – Launch Repackager

AdminStudio application was opened and the Repackager tool was launched.
The option “Capture an Installation Using Repackaging Wizard” was selected to start the packaging process.



Step 2 – Select Repackaging Method

In the Repackaging Wizard, the method selection screen appeared.
The option “Installation Monitoring (Recommended)” was selected.
This method monitors all system changes during the installation of an application.



Step 3 – Enter Product Information

The product information for the application was entered.
Details entered:
Program File:
Copy code

Notepad++ Installer
Product Name:
Notepad++

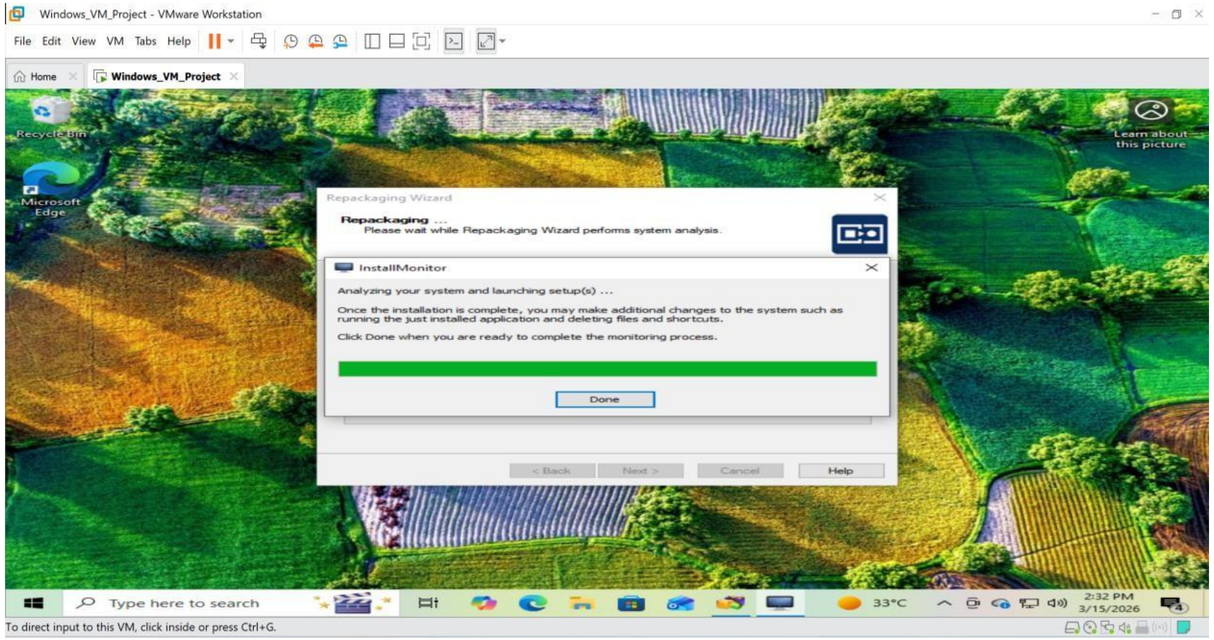
Company Name:
Student Project

Version:
1.0

Step 4 – Pre-Installation System Scan

The Repackaging Wizard performed a pre-installation scan of the system.

This scan captured the current state of the system including files, registry entries, and shortcuts.



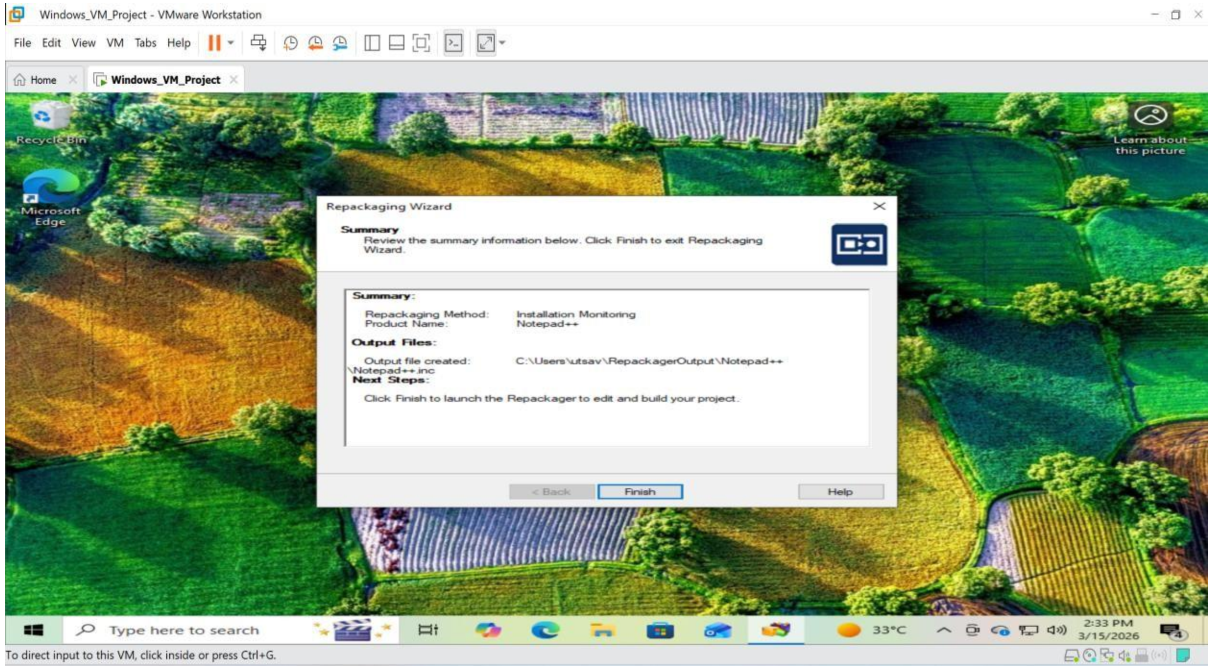
Step 5 – Application Installation

After the pre-scan, the selected application (Notepad++) installer was executed

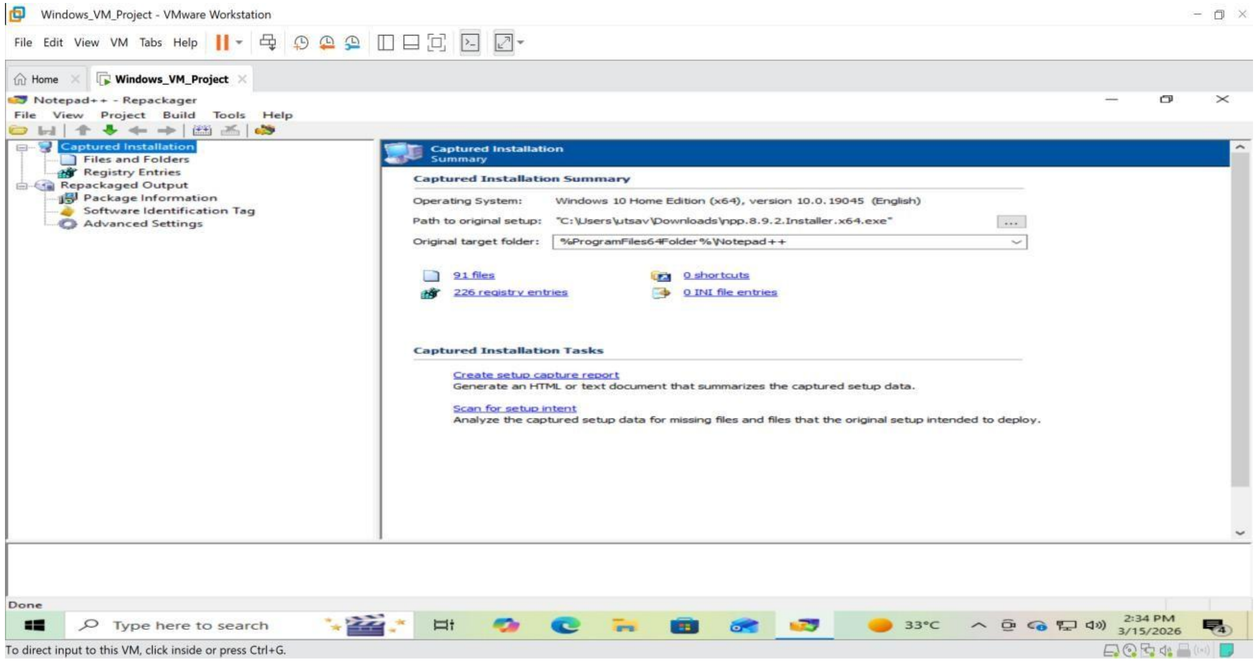
.

The application was installed normally using the setup wizard.
Steps performed:

Click Next
Click Install
Complete installation
Click Finish



Step 6 – Post-Installation Scan



After the application installation was completed, the Repackaging Wizard performed a post-installation scan. This scan compared the system state before and after installation to identify:

New files created

Registry entries added

Shortcuts generated

Step 7 – Captured Installation Summary

The wizard displayed the Captured Installation Summary which showed the changes detected during installation.

Example captured data:

Files captured:

91 files

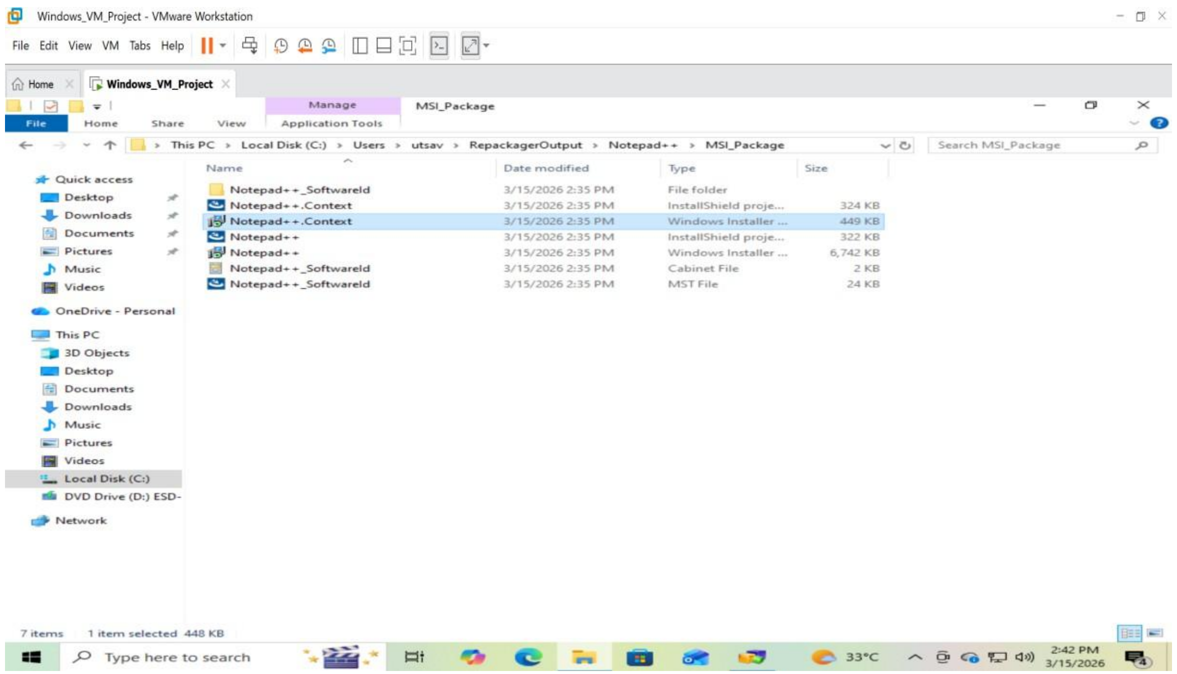
Registry entries captured:

226 registry entries

Step 8 – Build MSI Package

The captured installation data was then converted into an MSI package using the Build option.
The packaging tool generated the installer files in the output folder.
Output directory:

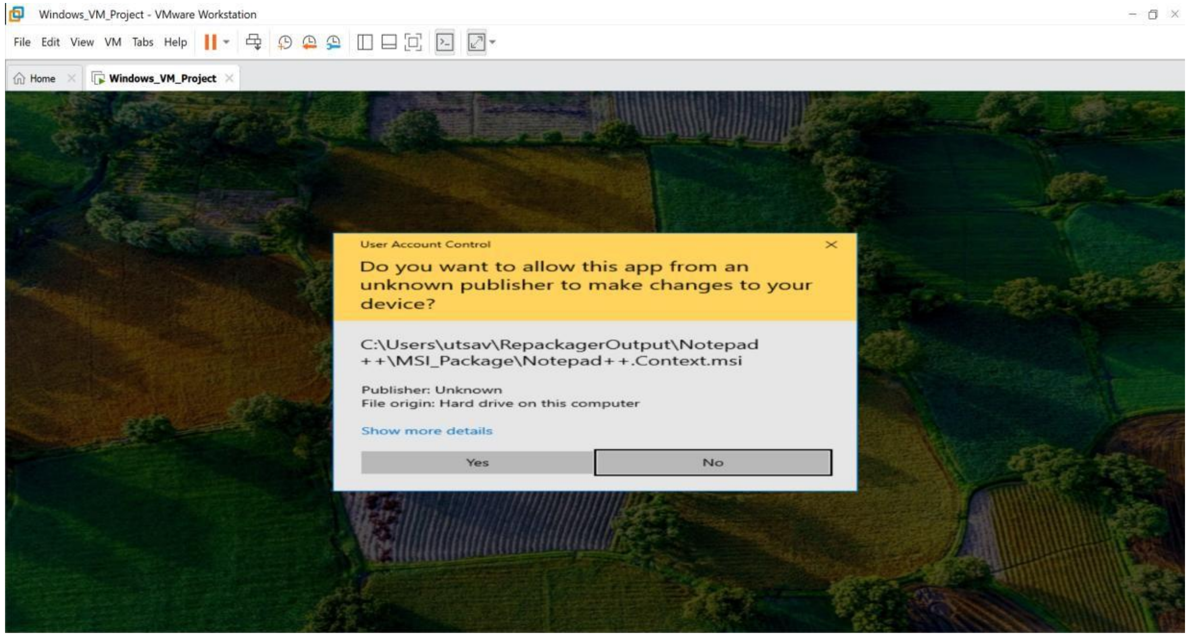
C:\Users\utsav\RepackagerOutput



Step 9 – Generated MSI Package

The final MSI installer package was generated in the MSI_Package folder.
Example installer file:

Notepad++ Windows Installer Package
This MSI package can now be used to deploy the application automatically.

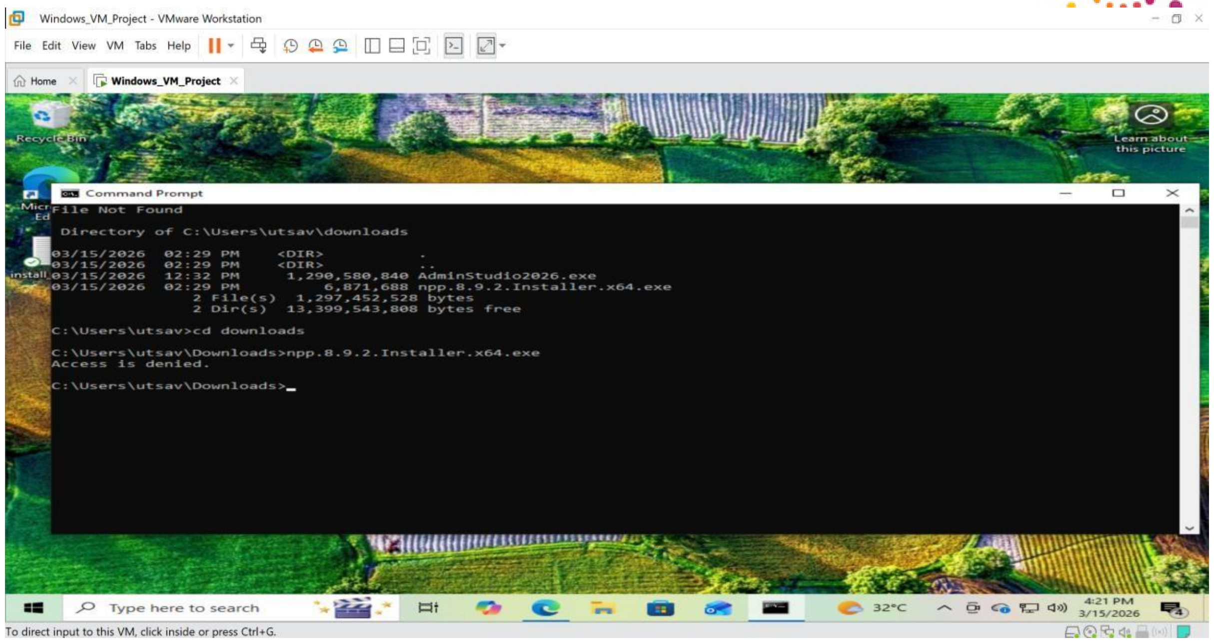


Create a Silent Installation Script

Step 1 – Download the Application Installer

First, the installer file of Notepad++ was downloaded from the official website and saved in the Downloads folder of the virtual machine.
The installer file used for this task is:

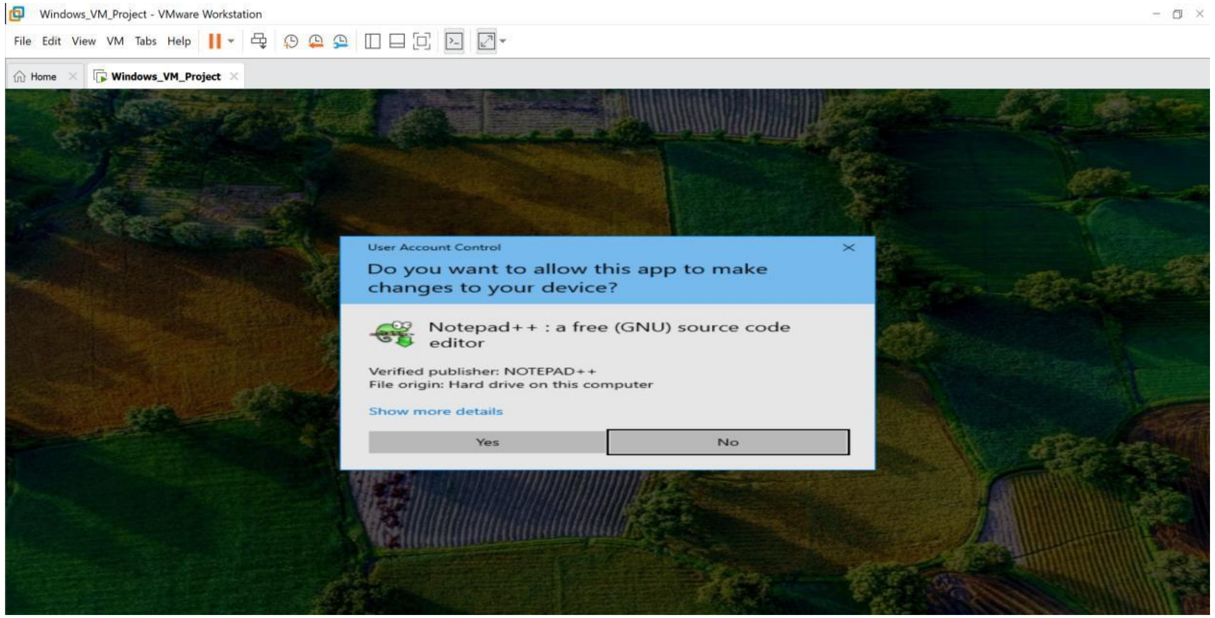
npp.8.9.2.Installer.x64.exe



Step 2 – Identify Silent Installation Command

The Notepad++ installer supports silent installation using the /S switch.
Silent installation command:

npp.8.9.2.Installer.x64.exe /S
This command installs the application silently without displaying the installation wizard.

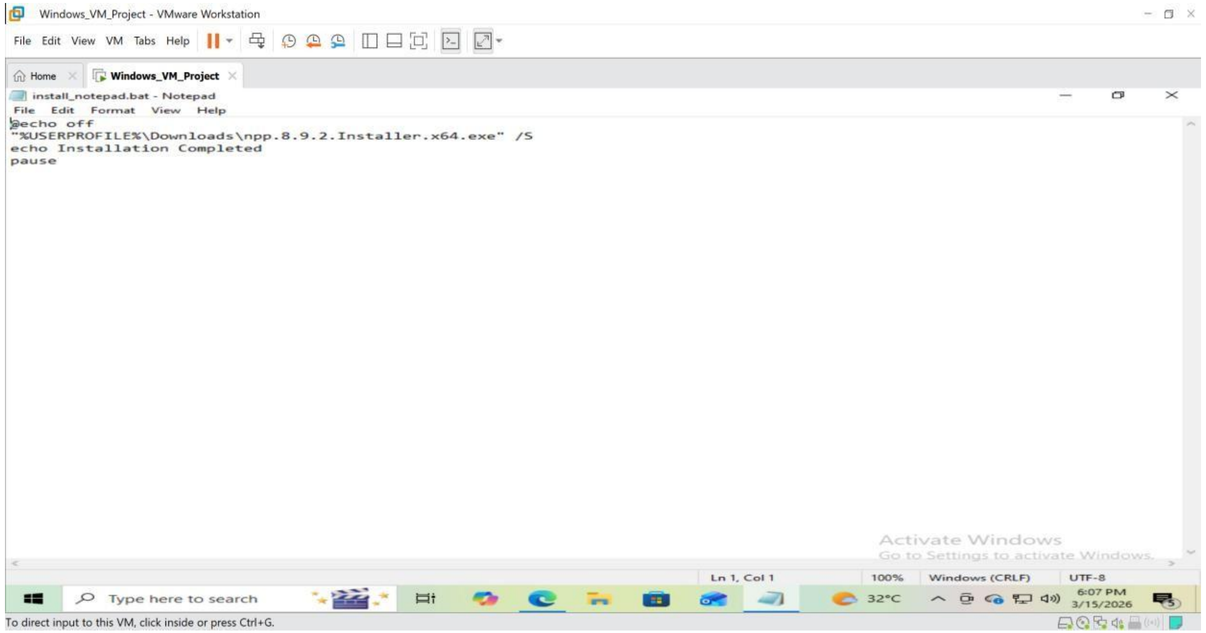


Step 3 – Create a Batch Script

To automate the installation process, a batch script (.bat file) was created.
File name:
install_notepad.bat
Batch script code:

```
@echo off
npp.8.9.2.Installer.x64.exe /S
echo Installation Completed
pause
```

This script runs the installer using the silent installation switch.



Step 4 – Execute the Script

The batch script was executed through Command Prompt in the virtual machine.
Command used:

install_notepad.bat
The script installed the application automatically without user interaction.

Step 5 – Verification

After executing the script, the installation was verified by launching Notepad++ from the Start Menu.
The application opened successfully, confirming that the silent installation worked correctly.

Unit 6 :Modify MSI Properties Using Orca

1.Steps:-Download and install Orca tool from the Microsoft edge.

2.Steps:-Open Orca from the Start Menu.

3.Steps:-Click on file ⍰ Open

4.Steps:-Browser and select the .msi file(7zip.msi).

5.Steps:-In the left panel , select the property table.

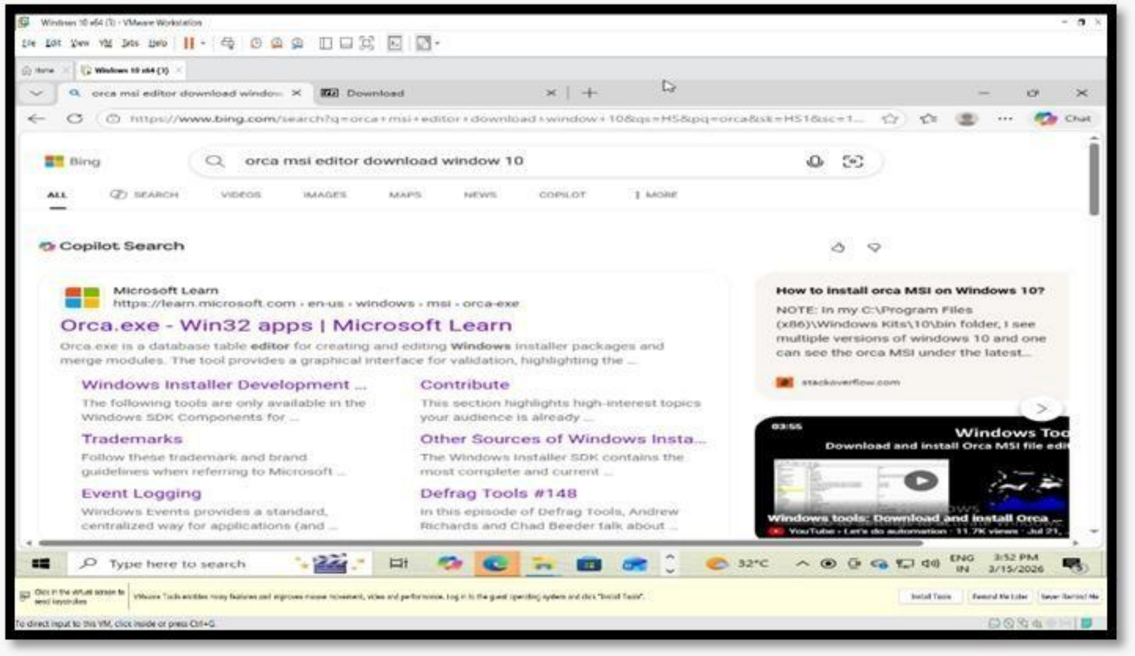
6.Steps:-Modify properties such as:

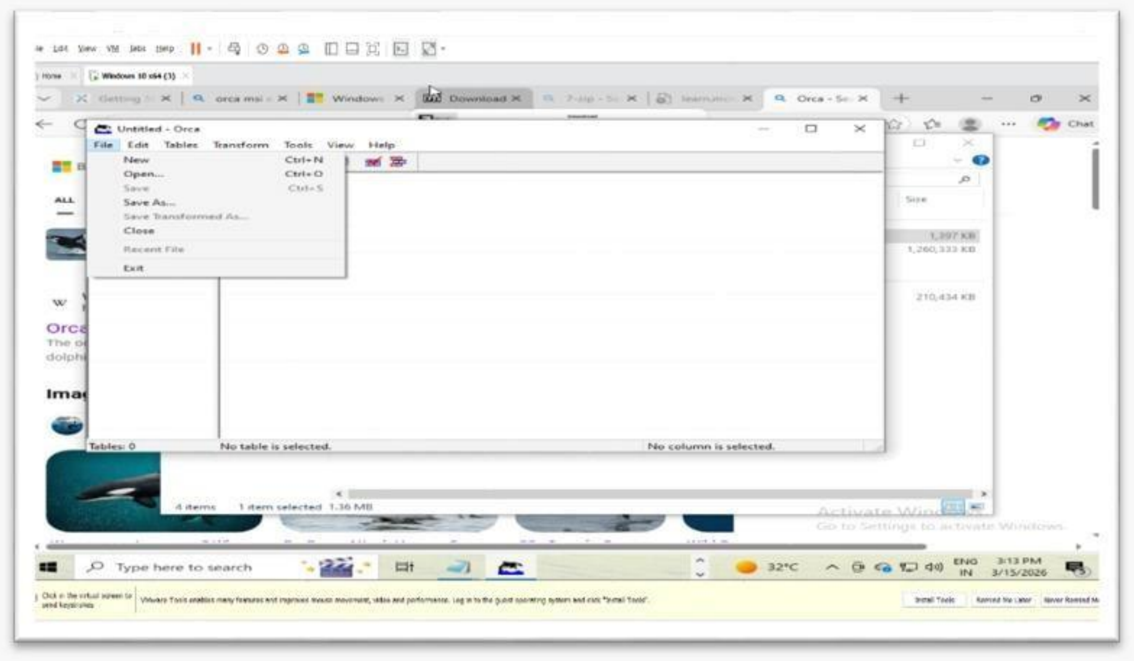
ProductName

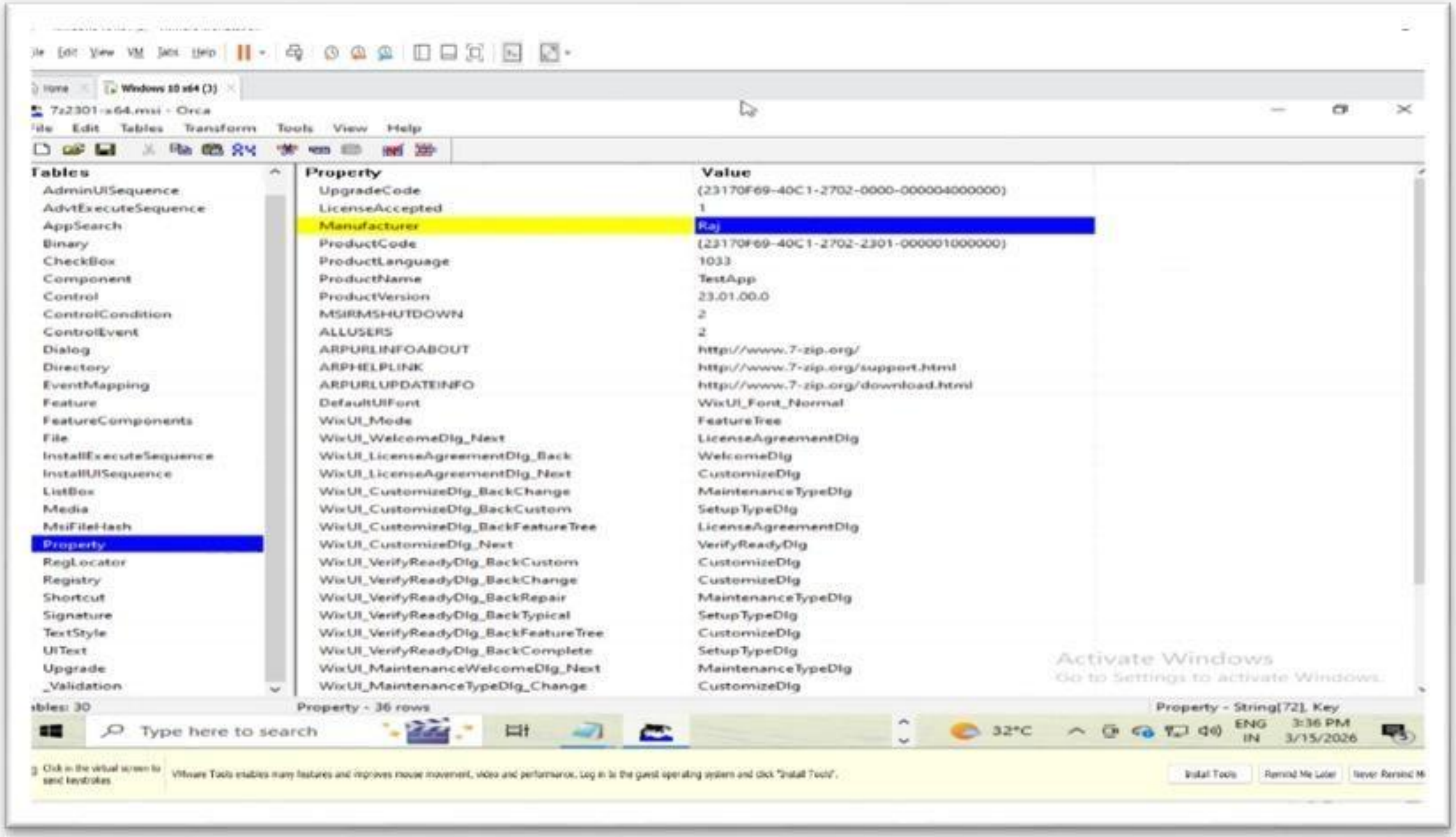
Manufacturer

7.Steps:-Click File ⍰ Save As and save the modified MSI file.

8.Steps:-Run the modified .msi file to verify the changes.



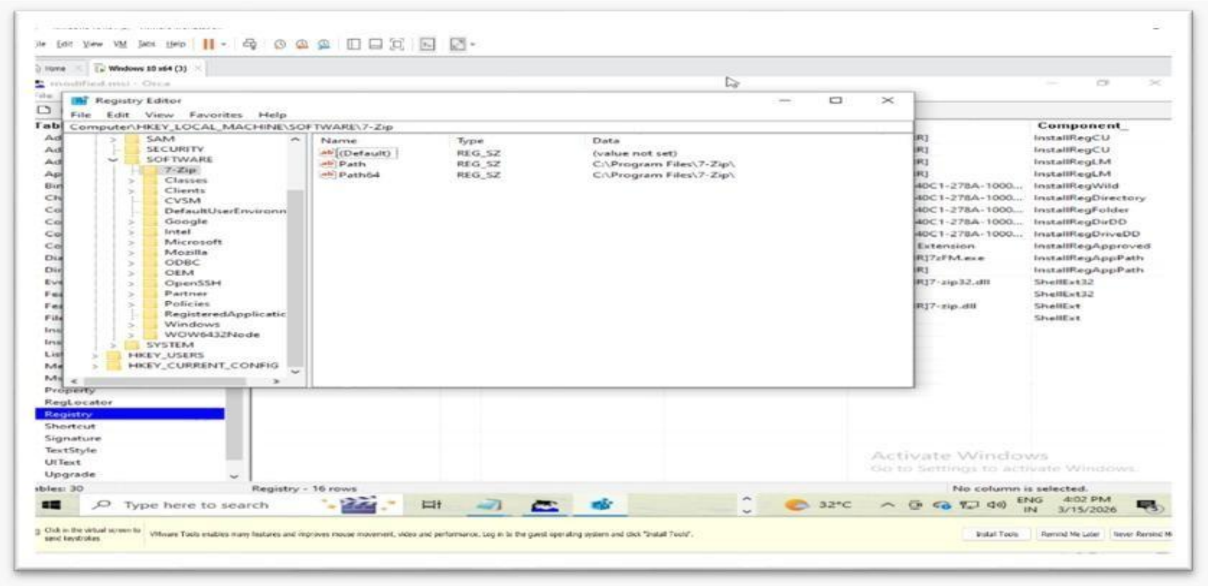


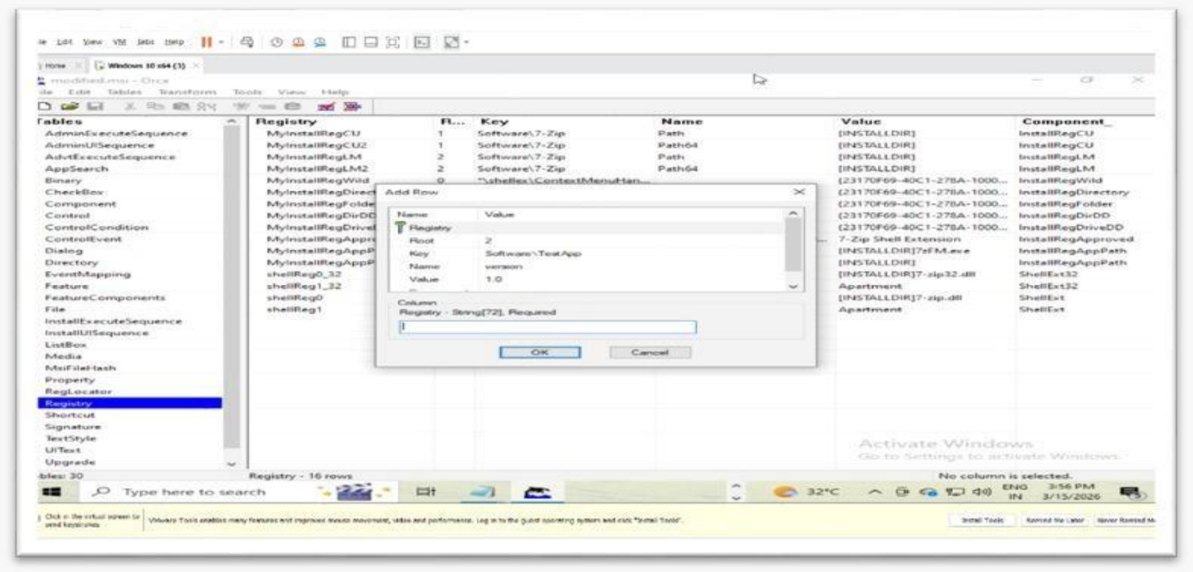

7z2301-x64.msi - Orca
File Edit Tables Transform Tools View Help
Tables
AdminUISequence
AdvtExecuteSequence
AppSearch
Binary
CheckBox
Component
Control
ControlCondition
ControlEvent
Dialog
Directory
EventMapping
Feature
FeatureComponents
File
InstallExecuteSequence
InstallUISequence
ListBox
Media
MsiFileHash
Property
RegLocator
Registry
Shortcut
Signature
TextStyle
UIText
Upgrade
_Validation
Property | Value
UpgradeCode | {23170F69-40C1-2702-0000-000004000000}
LicenseAccepted | 1
Manufacturer | Raj
ProductCode | {23170F69-40C1-2702-2301-000001000000}
ProductLanguage | 1033
ProductName | TestApp
ProductVersion | 23.01.00.0
MSIRMSHUTDOWN | 2
ALLUSERS | 2
ARPURLINFOABOUT | http://www.7-zip.org/
ARPHELPLINK | http://www.7-zip.org/support.html
ARPURLUPDATEINFO | http://www.7-zip.org/download.html
DefaultUIFont | WixUI_Font_Normal
WixUI_Mode | FeatureTree
WixUI_WelcomeDlg_Next | LicenseAgreementDlg
WixUI_LicenseAgreementDlg_Back | WelcomeDlg
WixUI_LicenseAgreementDlg_Next | CustomizeDlg
WixUI_CustomizeDlg_BackChange | MaintenanceTypeDlg
WixUI_CustomizeDlg_BackCustom | SetupTypeDlg
WixUI_CustomizeDlg_BackFeatureTree | LicenseAgreementDlg
WixUI_CustomizeDlg_Next | VerifyReadyDlg
WixUI_VerifyReadyDlg_BackCustom | CustomizeDlg
WixUI_VerifyReadyDlg_BackChange | CustomizeDlg
WixUI_VerifyReadyDlg_BackRepair | MaintenanceTypeDlg
WixUI_VerifyReadyDlg_BackTypical | SetupTypeDlg
WixUI_VerifyReadyDlg_BackFeatureTree | CustomizeDlg
WixUI_VerifyReadyDlg_BackComplete | SetupTypeDlg
WixUI_MaintenanceWelcomeDlg_Next | MaintenanceTypeDlg
WixUI_MaintenanceTypeDlg_Change | CustomizeDlg
Activate Windows
Go to Settings to activate Windows.
Tables: 30
Property - 36 rows
Property - String[72], Key
Type here to search
32°C
ENG IN
3:36 PM 3/15/2026

Unit 7:-Add Custom Registry Entries

Steps:-

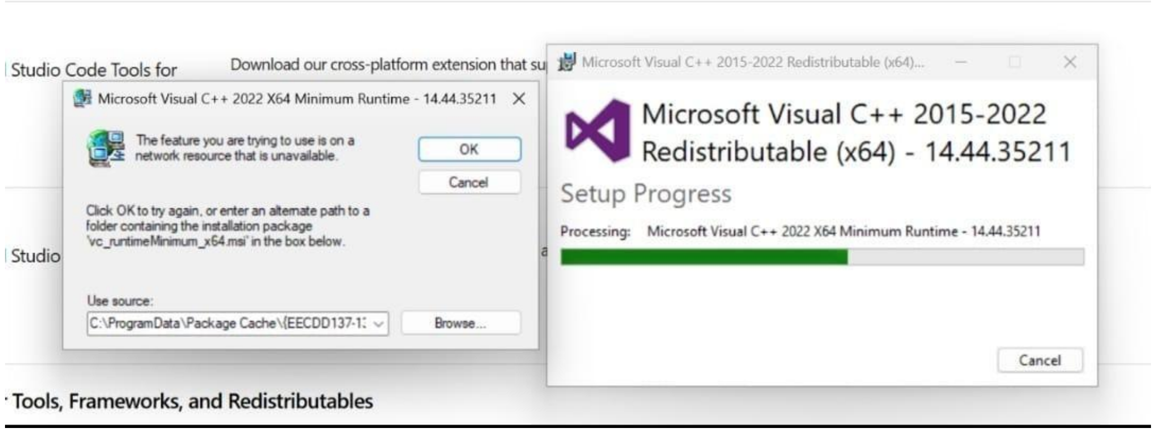
Open Orca Tool from the Start Menu.
Click File → Open and select the MSI file.
In the left panel, select the Registry table.
Right-click and choose Add Row.
Enter registry details (Root, Key, Name, Value).
Click File → Save As to save the modified MSI.
Run the MSI installer.
Open Registry Editor to verify.





UNIT 8 – Package an Application with Prerequisites

Objective
The objective of this task is to package an application along with its required prerequisites. Some applications require additional software components such as runtime libraries or frameworks before they can function properly. These prerequisites must be installed before installing the main application to ensure smooth operation.



Step 1 – Identify Application Prerequisites

Before packaging an application, it is important to identify the prerequisites required for the software to run properly.
For this project, the selected application is Notepad++.
Although Notepad++ is lightweight, many enterprise applications typically require components such as:
Microsoft Visual C++ Redistributable
.NET Framework
Windows Installer Components
In this example, Microsoft Visual C++ Redistributable was selected as the prerequisite component.

Step 2 – Download the Prerequisite Package.

The prerequisite installer package was downloaded from the official Microsoft website.
The downloaded file name is:
vc_redist.x64.exe
This file was saved in the Downloads folder of the virtual machine.

Step 3 – Install the Prerequisite SilentlyThe prerequisite package was installed silently using the command prompt.

The following command was used to install the prerequisite:

vc_redist.x64.exe /install /quiet /norestartExplanation of parameters:/install – installs the Visual C++ Redistributable/quiet – performs the installation silently without user interaction/norestart – prevents the system from restarting automaticallyThis ensures the prerequisite component is installed before deploying the main application.

Step 4 – Install the Main ApplicationAfter installing the prerequisite, the main application (Notepad++) was installed using a silent installation command.

The following command was used:

npp.8.9.2.Installer.x64.exe /SThe /S switch installs the application silently without displaying the installation wizard.

Step 5 – VerificationAfter installing both the prerequisite and the main application, the installation was verified

.The Notepad++ application was launched from the Start Menu to confirm that the application was installed successfully and working properly.

Unit – 9: Create a Transform (.MST) File

Objective

The objective of this unit is to create a Transform file (.MST) that allows customization of the MSI installation package without modifying the original MSI file. A transform file is used to apply specific configuration changes such as installation settings, features, or properties during deployment.

Step 1: Open the MSI File in Orca

The first step was to open the MSI installation package using the Orca MSI Editor.
Steps performed:
1. Launch the Orca Tool
2. Click File → Open
3. Select the required MSI file Once the file was opened, the installer database tables became visible inside the Orca interface.

Step 2: Create a New Transform

After opening the MSI file, a new transform file was created.
Steps performed:

1. Click Transform in the menu bar
2. Select New Transform

This allows modifications to be recorded separately without changing the original MSI file.

Step 3: Modify InstallationProperties

Some installation properties were modified according to the project requirements. Examples of modifications include:

- Changing the Product Name
- Editing the Manufacturer name
- Updating the installation directory

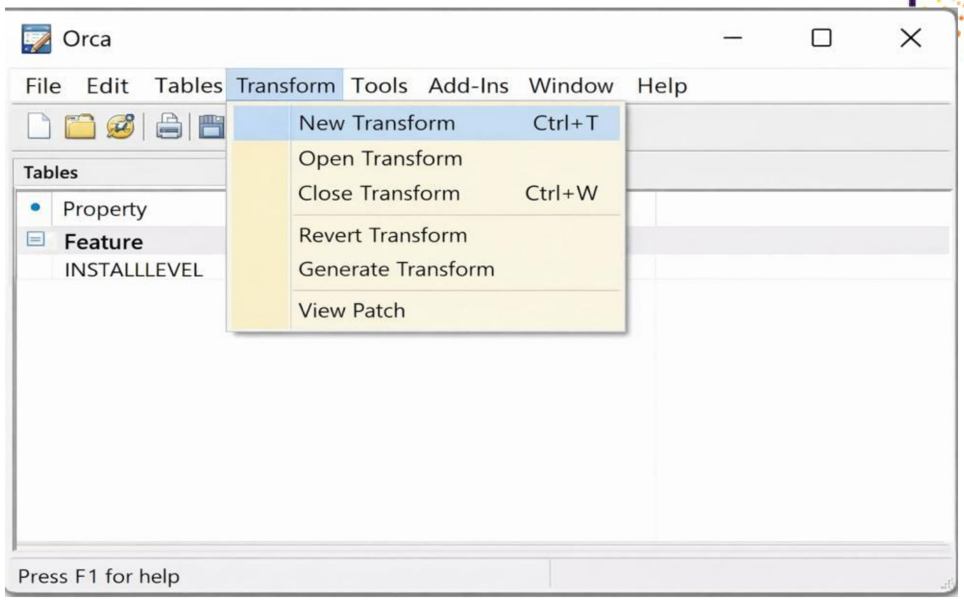
These changes help customize the installation process.

Step 4: Generate the Transform File

After completing the required modifications, the transform file was generated.

Steps performed:

1. Click Transform → Generate Transform
2. 2. Save the file with the extension .MST Example file name: custom_transform.mst

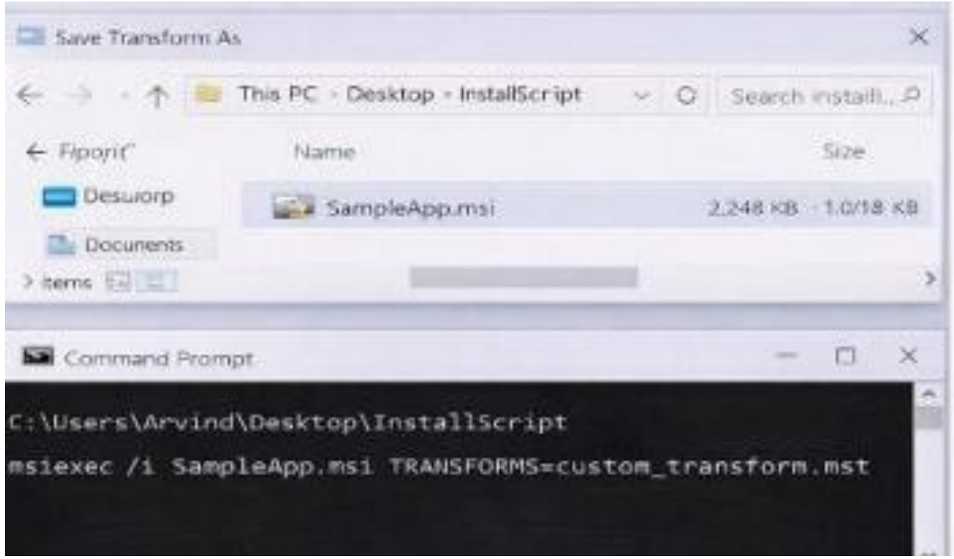


Step 5: Deploy MSI with Transform

The MSI package was installed using the transform file.

Example command used: msiexec /i application.msi TRANSFORMS=custom_transform.mst

This command installs the MSI package while applying the modifications stored in the transform file.



Unit – 10: Implement Version Control

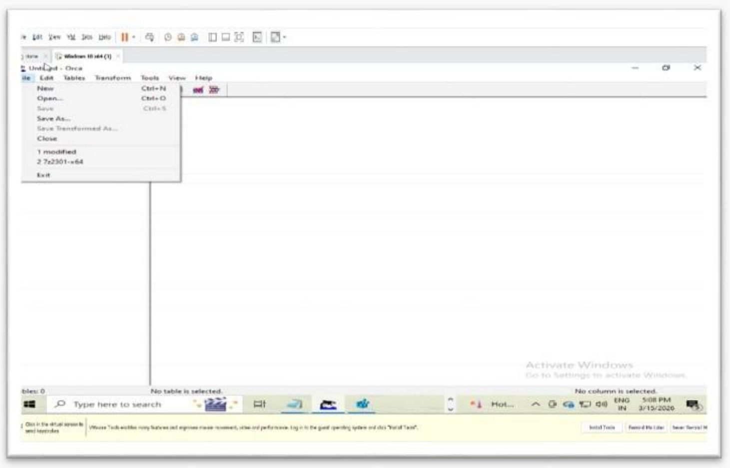
Objective :

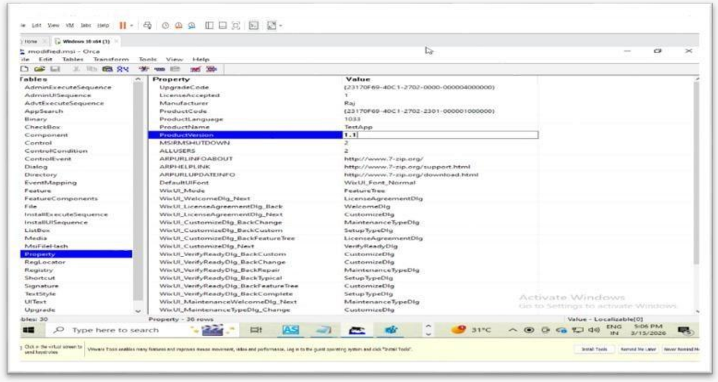
The objective of this unit is to implement version control for the MSI packages created during the application packaging process. Version control helps in tracking changes, maintaining different versions of the package, and ensuring that previous versions are available if rollback or troubleshooting is required.

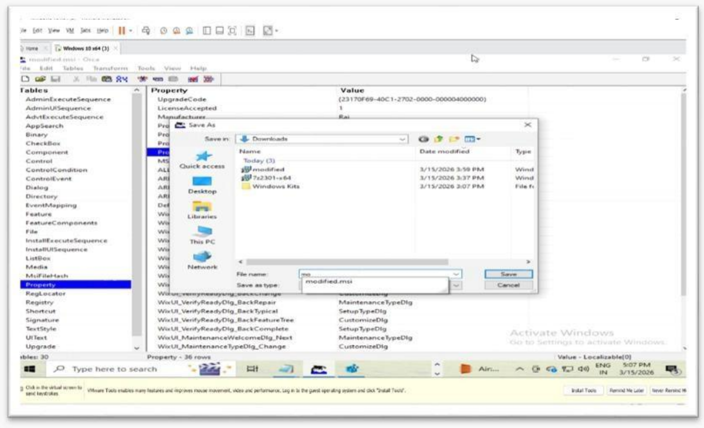
Steps of doing this:

Step 1: Assign Version Number

- Each MSI package was assigned a version number to identify the release of the application package. The version number was updated in the Product Version property of the MSI package using the Orca tool or packaging tool. Example version format: 1.0.0







Step 2: Maintain Version Log

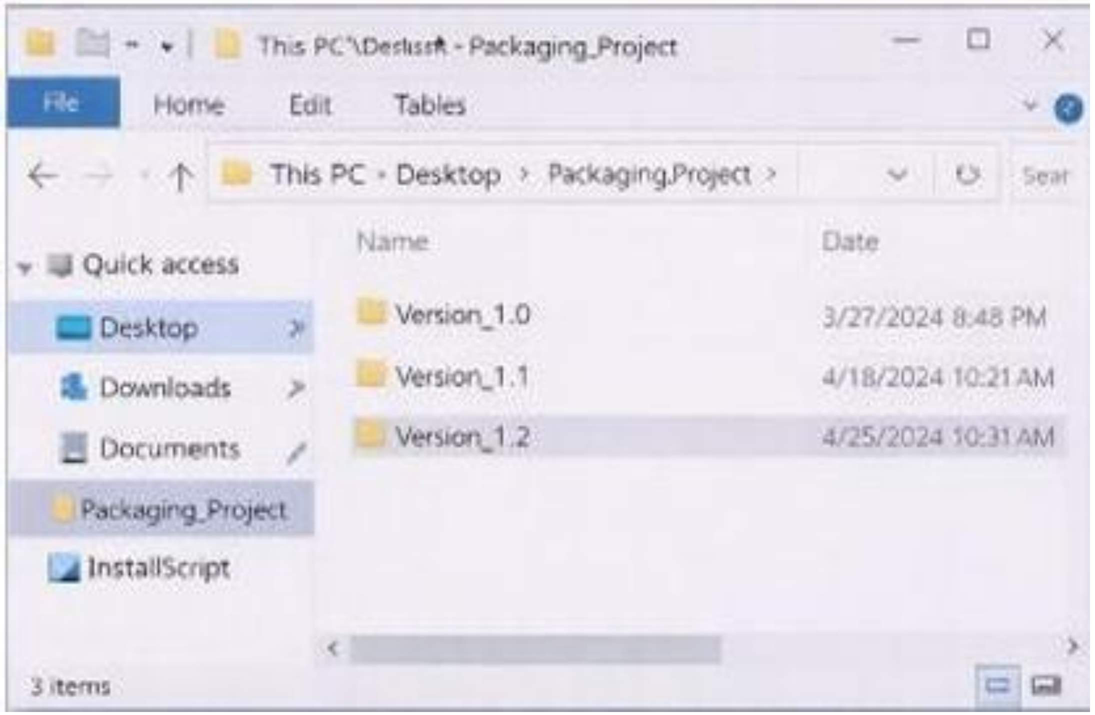
A version log or changelog file was created to record the modifications made to the MSI package. The log included information such as:

- Version number
- Date of modification
- Changes applied to the package
- Author or packager name

Step 3: Store Previous Versions

Older versions of the MSI package were stored in a secure folder or repository. This ensures that:

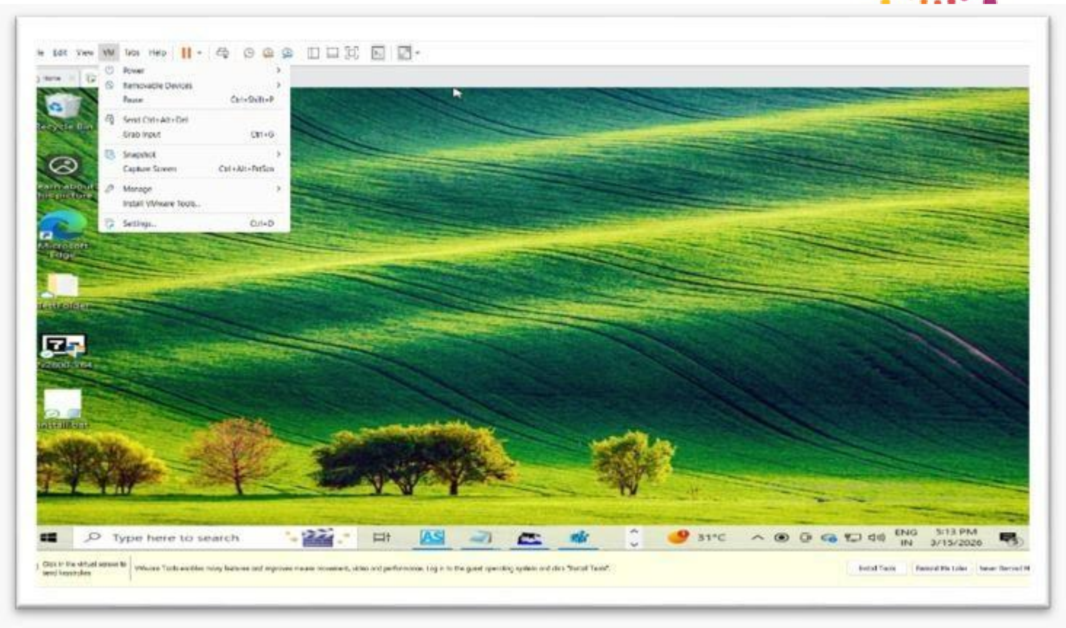
- Previous versions can be restored if needed
- Troubleshooting can be done easily
- Deployment history is maintained



Step 4: Verify Version Control

The MSI package was checked to confirm that the updated version number appeared correctly during installation and inside the application details. This confirmed that the version control process was implemented successfully.

Unit – 11: Test Package Installation on a Clean Virtual Machine



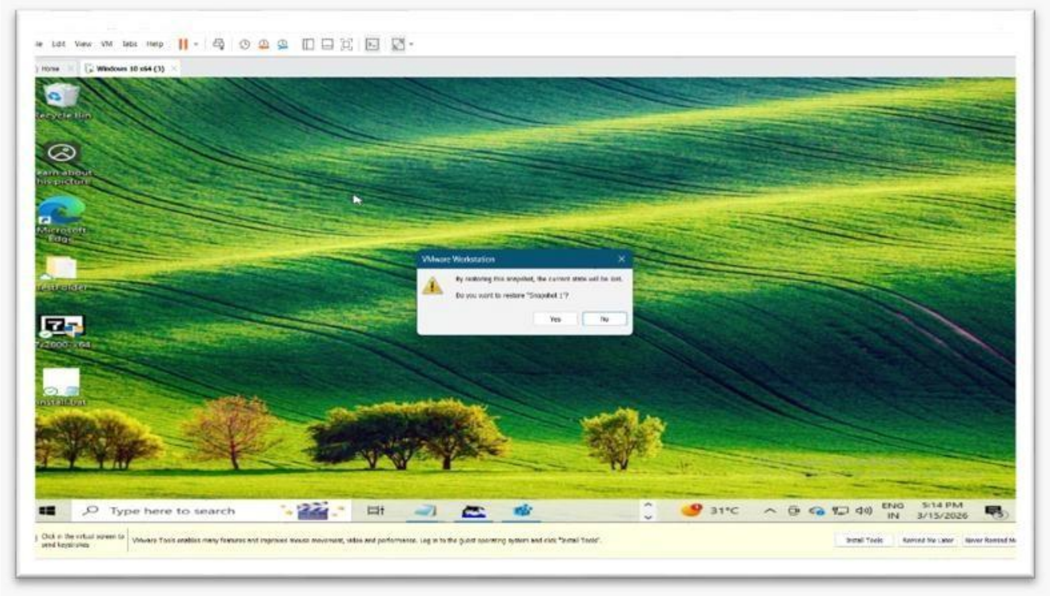
Objective

The objective of this unit is to test the installation of the packaged MSI application on a clean virtual machine. This ensures that the application installs correctly and functions properly without any errors or missing dependencies

Steps to do this:

Step 1: Restore Clean Snapshot Before testing the package, the virtual machine was reverted to the clean snapshot that was created earlier.
Steps performed:

1. Open VMware Workstation
2. 2. Select the virtual machine
3. 3. Click VM → Snapshot → Revert to Snapshot



This restored the system to its clean state before any application installation.

Step 2: Copy the MSI Package

After restoring the snapshot, the MSI package was copied into the virtual machine.
The installer file was placed in a convenient location such as the Desktop or a project folder.

Example file: SampleApp.msi

Step 3: Install the MSI Package

The MSI installer was executed to test the installation process. Steps performed:

1. Double-click the MSI file
2. 2. Follow the installation wizard
3. 3. Complete the installation process Alternatively,

the package can also be installed using command prompt.

Example command: msiexec /i SampleApp.msi

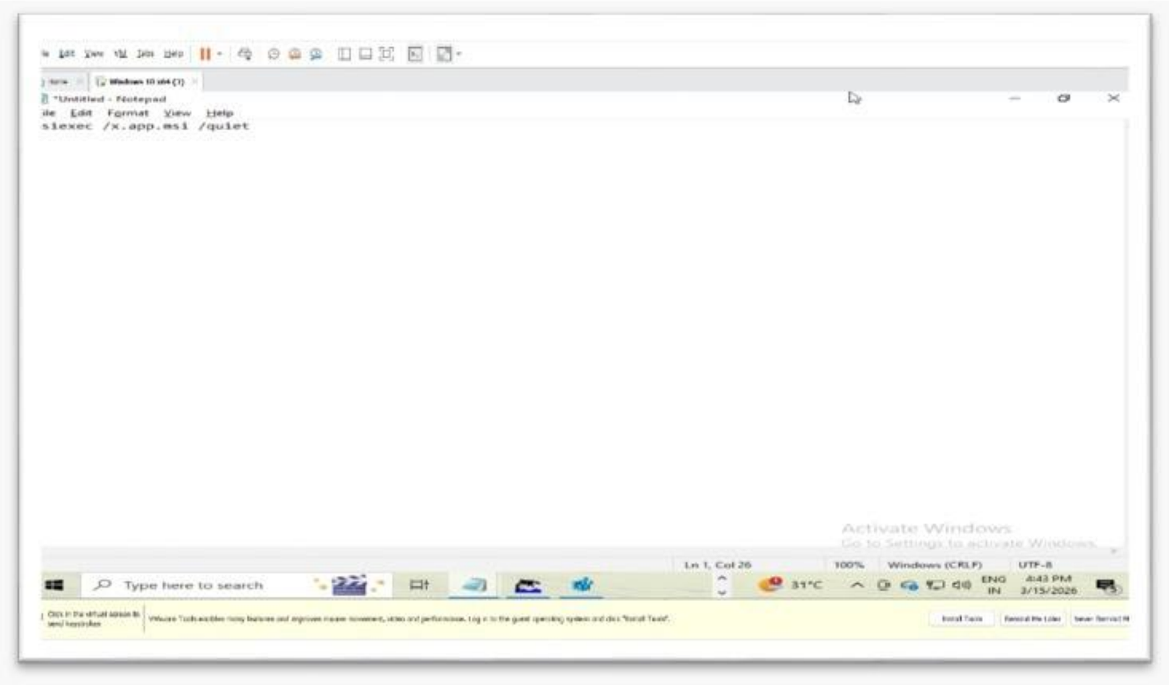
Step 4: Verify Application Functionality

After installation, the application was launched to verify that it works correctly.
The following checks were performed:

- Application opened successfully
- Installation files were placed in the correct directory
- Registry entries were created properly
- Shortcuts appeared in the Start Menu

Unit – 12: Create Uninstall Script

Objective:
The objective of this unit is to create an uninstall script that removes the installed application from the system automatically. This script allows administrators to uninstall the application silently without any user interaction.



Steps to do this:-

Step 1: Identify the Uninstall Command

MSI applications can be removed using the **msiexec command** with uninstall parameters.
Example uninstall command:

msiexec /x SampleApp.msi /quiet Explanation:

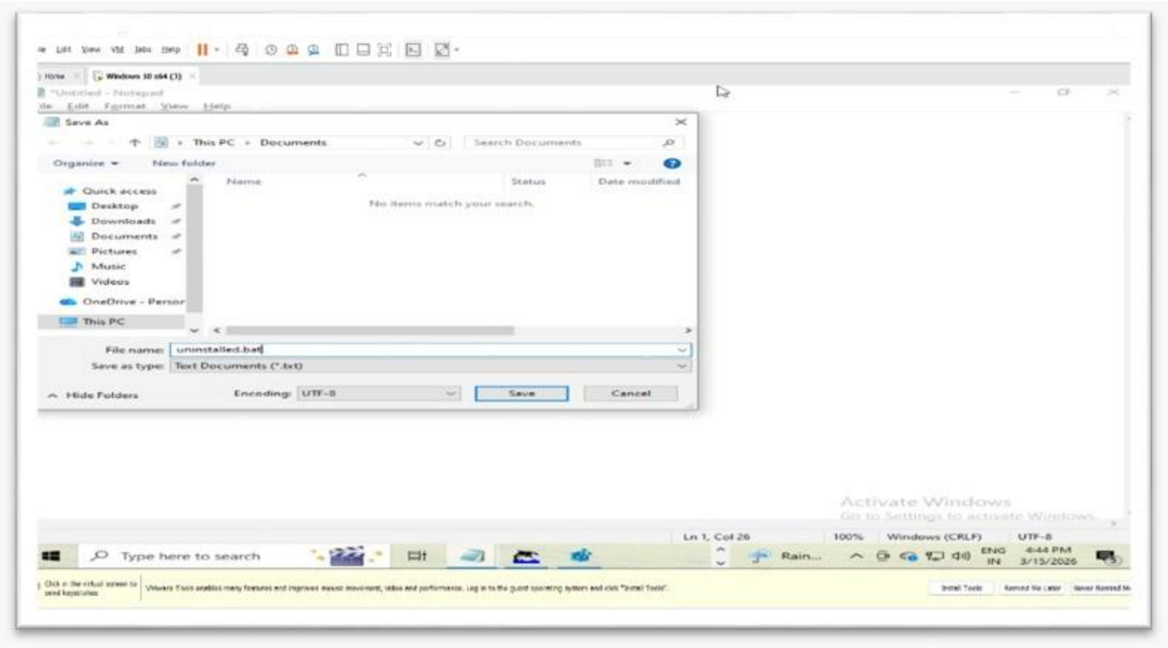
- /x → Uninstall the application
- /quiet → Runs uninstall silently without showing the interface

Step 2: Create an Uninstall Script

To automate the uninstall process, a batch script was created.

Steps performed:

1. Open Notepad
2. 2. Write the uninstall command
3. 3. Save the file as a batch script Example file name: uninstall.bat



Step 3: Run the Uninstall Script

The uninstall script was executed to remove the application from the system.

Steps performed:

1. Open Command Prompt
2. Navigate to the folder where the script is saved
3. Run the script Example command:

uninstall.bat The uninstall process started and removed the application silently.

Step 4: Verify Application Removal

After running the uninstall script, the system was checked to confirm that the application was completely removed.

Verification steps included:

- Checking Apps & Features in Windows settings
- Confirming that application files were removed
- Ensuring related registry entries were deleted